**Theorem 6.2.1: Subspace Test**

*A subset $U$ of a vector space is a subspace of $V$ if and only if it satisfies the following three conditions:*

1. *$\mathbf{0}$ lies in $U$ where $\mathbf{0}$ is the zero vector of $V$.*
2. *If $\mathbf{u}_1$ and $\mathbf{u}_2$ are in $U$, then $\mathbf{u}_1 + \mathbf{u}_2$ is also in $U$.*
3. *If $\mathbf{u}$ is in $U$, then $a\mathbf{u}$ is also in $U$ for each scalar $a$.*

Proof. If $U$ is a subspace of $V$, then (2) and (3) hold by axioms A1 and S1 respectively, applied to the vector space $U$. Since $U$ is nonempty (it is a vector space), choose $\mathbf{u}$ in $U$. Then (1) holds because $\mathbf{0} = 0\mathbf{u}$ is in $U$ by (3) and Theorem 6.1.3.

Conversely, if (1), (2), and (3) hold, then axioms A1 and S1 hold because of (2) and (3), and axioms A2, A3, S2, S3, S4, and S5 hold in $U$ because they hold in $V$. Axiom A4 holds because the zero vector $\mathbf{0}$ of $V$ is actually in $U$ by (1), and so serves as the zero of $U$. Finally, given $\mathbf{u}$ in $U$, then its negative $-\mathbf{u}$ in $V$ is again in $U$ by (3) because $-\mathbf{u} = (-1)\mathbf{u}$ (again using Theorem 6.1.3). Hence $-\mathbf{u}$ serves as the negative of $\mathbf{u}$ in $U$. □

Note that the proof of Theorem 6.2.1 shows that if $U$ is a subspace of $V$, then $U$ and $V$ share the same zero vector, and that the negative of a vector in the space $U$ is the same as its negative in $V$.

**Example 6.2.1**

If $V$ is any vector space, show that $\{\mathbf{0}\}$ and $V$ are subspaces of $V$.

Solution. $U = V$ clearly satisfies the conditions of the subspace test. As to $U = \{\mathbf{0}\}$, it satisfies the conditions because $\mathbf{0}+\mathbf{0} = \mathbf{0}$ and $a\mathbf{0} = \mathbf{0}$ for all $a$ in $\mathbb{R}$.

The vector space $\{\mathbf{0}\}$ is called the **zero subspace** of $V$.

**Example 6.2.2**

Let $\mathbf{v}$ be a vector in a vector space $V$. Show that the set

$$\mathbb{R}\mathbf{v} = \{a\mathbf{v} \mid a \text{ in } \mathbb{R}\}$$

of all scalar multiples of $\mathbf{v}$ is a subspace of $V$.

Solution. Because $\mathbf{0} = 0\mathbf{v}$, it is clear that $\mathbf{0}$ lies in $\mathbb{R}\mathbf{v}$. Given two vectors $a\mathbf{v}$ and $a_1\mathbf{v}$ in $\mathbb{R}\mathbf{v}$, their sum $a\mathbf{v}+a_1\mathbf{v} = (a+a_1)\mathbf{v}$ is also a scalar multiple of $\mathbf{v}$ and so lies in $\mathbb{R}\mathbf{v}$. Hence $\mathbb{R}\mathbf{v}$ is closed under addition. Finally, given $a\mathbf{v}$, $r(a\mathbf{v}) = (ra)\mathbf{v}$ lies in $\mathbb{R}\mathbf{v}$ for all $r \in \mathbb{R}$, so $\mathbb{R}\mathbf{v}$ is closed under scalar multiplication. Hence the subspace test applies.

In particular, given $\mathbf{d} \neq \mathbf{0}$ in $\mathbb{R}^3$, $\mathbb{R}\mathbf{d}$ is the line through the origin with direction vector $\mathbf{d}$.

The space $\mathbb{R}\mathbf{v}$ in Example 6.2.2 is described by giving the *form* of each vector in $\mathbb{R}\mathbf{v}$. The next example describes a subset $U$ of the space $\mathbf{M}_{nn}$ by giving a *condition* that each matrix of $U$ must satisfy.

**Example 6.2.3**

Let $A$ be a fixed matrix in $\mathbf{M}_{nn}$. Show that $U = \{X \text{ in } \mathbf{M}_{nn} \mid AX = XA\}$ is a subspace of $\mathbf{M}_{nn}$.

Solution. If 0 is the $n \times n$ zero matrix, then $A0 = 0A$, so 0 satisfies the condition for membership in $U$. Next suppose that $X$ and $X_1$ lie in $U$ so that $AX = XA$ and $AX_1 = X_1A$. Then

$$A(X+X_1) = AX + AX_1 = XA + X_1A + (X+X_1)A$$
$$A(aX) = a(AX) = a(XA) = (aX)A$$

for all $a$ in $\mathbb{R}$, so both $X + X_1$ and $aX$ lie in $U$. Hence $U$ is a subspace of $\mathbf{M}_{nn}$.

Suppose $p(x)$ is a polynomial and $a$ is a number. Then the number $p(a)$ obtained by replacing $x$ by $a$ in the expression for $p(x)$ is called the **evaluation** of $p(x)$ at $a$. For example, if $p(x) = 5 - 6x + 2x^2$, then the evaluation of $p(x)$ at $a = 2$ is $p(2) = 5 - 12 + 8 = 1$. If $p(a) = 0$, the number $a$ is called a **root** of $p(x)$.

**Example 6.2.4**

Consider the set $U$ of all polynomials in $\mathbf{P}$ that have 3 as a root:

$$U = \{p(x) \in \mathbf{P} \mid p(3) = 0\}$$

Show that $U$ is a subspace of $\mathbf{P}$.

Solution. Clearly, the zero polynomial lies in $U$. Now let $p(x)$ and $q(x)$ lie in $U$ so $p(3) = 0$ and $q(3) = 0$. We have $(p+q)(x) = p(x) + q(x)$ for all $x$, so $(p+q)(3) = p(3) + q(3) = 0 + 0 = 0$, and $U$ is closed under addition. The verification that $U$ is closed under scalar multiplication is similar.

Recall that the space $\mathbf{P}_n$ consists of all polynomials of the form

$$a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

where $a_0, a_1, a_2, \ldots, a_n$ are real numbers, and so is closed under the addition and scalar multiplication in $\mathbf{P}$. Moreover, the zero polynomial is included in $\mathbf{P}_n$. Thus the subspace test gives Example 6.2.5.

**Example 6.2.5**

$\mathbf{P}_n$ is a subspace of $\mathbf{P}$ for each $n \geq 0$.

The next example involves the notion of the derivative $f'$ of a function $f$. (If the reader is not familiar with calculus, this example may be omitted.) A function $f$ defined on the interval $[a, b]$ is called **differentiable** if the derivative $f'(r)$ exists at every $r$ in $[a, b]$.

**Example 6.2.6**

Show that the subset $\mathbf{D}[a, b]$ of all **differentiable functions** on $[a, b]$ is a subspace of the vector space $\mathbf{F}[a, b]$ of all functions on $[a, b]$.

Solution. The derivative of any constant function is the constant function 0; in particular, 0 itself is differentiable and so lies in $\mathbf{D}[a, b]$. If $f$ and $g$ both lie in $\mathbf{D}[a, b]$ (so that $f'$ and $g'$ exist), then it is a theorem of calculus that $f+g$ and $rf$ are both differentiable for any $r \in \mathbb{R}$. In fact, $(f+g)' = f' + g'$ and $(rf)' = rf'$, so both lie in $\mathbf{D}[a, b]$. This shows that $\mathbf{D}[a, b]$ is a subspace of $\mathbf{F}[a, b]$.

## Linear Combinations and Spanning Sets

**Definition 6.3 Linear Combinations and Spanning**

*Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a set of vectors in a vector space $V$. As in $\mathbb{R}^n$, a vector $\mathbf{v}$ is called a **linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ if it can be expressed in the form*

$$\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n$$

*where $a_1, a_2, \dots, a_n$ are scalars, called the **coefficients** of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. The set of all linear combinations of these vectors is called their **span**, and is denoted by*

$$\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n \mid a_i \text{ in } \mathbb{R}\}$$

If it happens that $V = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, these vectors are called a **spanning set** for $V$. For example, the span of two vectors $\mathbf{v}$ and $\mathbf{w}$ is the set

$$\operatorname{span}\{\mathbf{v}, \mathbf{w}\} = \{s\mathbf{v} + t\mathbf{w} \mid s \text{ and } t \text{ in } \mathbb{R}\}$$

of all sums of scalar multiples of these vectors.

**Example 6.2.7**

Consider the vectors $p_1 = 1 + x + 4x^2$ and $p_2 = 1 + 5x + x^2$ in $\mathbf{P}_2$. Determine whether $p_1$ and $p_2$ lie in $\operatorname{span}\{1 + 2x - x^2, 3 + 5x + 2x^2\}$.

Solution. For $p_1$, we want to determine if $s$ and $t$ exist such that

$$p_1 = s(1 + 2x - x^2) + t(3 + 5x + 2x^2)$$

Equating coefficients of powers of $x$ (where $x^0 = 1$) gives

$$1 = s + 3t, \quad 1 = 2s + 5t, \quad \text{and} \quad 4 = -s + 2t$$

These equations have the solution $s = -2$ and $t = 1$, so $p_1$ is indeed in $\operatorname{span}\{1 + 2x - x^2, 3 + 5x + 2x^2\}$.

Turning to $p_2 = 1+5x+x^2$, we are looking for $s$ and $t$ such that

$$p_2 = s(1+2x-x^2)+t(3+5x+2x^2)$$

Again equating coefficients of powers of $x$ gives equations $1 = s+3t$, $5 = 2s+5t$, and $1 = -s+2t$. But in this case there is no solution, so $p_2$ is *not* in $\text{span}\,\{1+2x-x^2,\ 3+5x+2x^2\}$.

We saw in Example 5.1.6 that $\mathbb{R}^m = \text{span}\,\{\mathbf{e}_1,\ \mathbf{e}_2,\ \dots,\ \mathbf{e}_m\}$ where the vectors $\mathbf{e}_1,\ \mathbf{e}_2,\ \dots,\ \mathbf{e}_m$ are the columns of the $m \times m$ identity matrix. Of course $\mathbb{R}^m = \mathbf{M}_{m1}$ is the set of all $m \times 1$ matrices, and there is an analogous spanning set for each space $\mathbf{M}_{mn}$. For example, each $2 \times 2$ matrix has the form

$$\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] = a\left[\begin{array}{cc} 1 & 0 \\ 0 & 0 \end{array}\right] + b\left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right] + c\left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array}\right] + d\left[\begin{array}{cc} 0 & 0 \\ 0 & 1 \end{array}\right]$$

so

$$\mathbf{M}_{22} = \text{span}\left\{\left[\begin{array}{cc} 1 & 0 \\ 0 & 0 \end{array}\right],\ \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right],\ \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array}\right],\ \left[\begin{array}{cc} 0 & 0 \\ 0 & 1 \end{array}\right]\right\}$$

Similarly, we obtain

**Example 6.2.8**

$\mathbf{M}_{mn}$ is the span of the set of all $m \times n$ matrices with exactly one entry equal to 1, and all other entries zero.

The fact that every polynomial in $\mathbf{P}_n$ has the form $a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ where each $a_i$ is in $\mathbb{R}$ shows that

**Example 6.2.9**

$\mathbf{P}_n = \text{span}\,\{1,\ x,\ x^2,\ \dots,\ x^n\}$.

In Example 6.2.2 we saw that $\text{span}\,\{\mathbf{v}\} = \{a\mathbf{v} \mid a \text{ in } \mathbb{R}\} = \mathbb{R}\mathbf{v}$ is a subspace for any vector $\mathbf{v}$ in a vector space $V$. More generally, the span of *any* set of vectors is a subspace. In fact, the proof of Theorem 5.1.1 goes through to prove:

**Theorem 6.2.2**

*Let $U = \text{span}\,\{\mathbf{v}_1,\ \mathbf{v}_2,\ \dots,\ \mathbf{v}_n\}$ in a vector space $V$. Then:*

1. *$U$ is a subspace of $V$ containing each of $\mathbf{v}_1,\ \mathbf{v}_2,\ \dots,\ \mathbf{v}_n$.*
2. *$U$ is the "smallest" subspace containing these vectors in the sense that any subspace that contains each of $\mathbf{v}_1,\ \mathbf{v}_2,\ \dots,\ \mathbf{v}_n$ must contain $U$.*

Here is how condition 2 in Theorem 6.2.2 is used. Given vectors $\mathbf{v}_1,\ \dots,\ \mathbf{v}_k$ in a vector space $V$ and a subspace $U \subseteq V$, then:

$$\text{span}\,\{\mathbf{v}_1,\ \dots,\ \mathbf{v}_n\} \subseteq U \Leftrightarrow \text{ each } \mathbf{v}_i \in U$$

The following examples illustrate this.

**Example 6.2.10**

Show that $\mathbf{P}_3 = \text{span}\,\{x^2+x^3,\ x,\ 2x^2+1,\ 3\}$.

Solution. Write $U = \text{span}\,\{x^2+x^3,\ x,\ 2x^2+1,\ 3\}$. Then $U \subseteq \mathbf{P}_3$, and we use the fact that $\mathbf{P}_3 = \text{span}\,\{1,\ x,\ x^2,\ x^3\}$ to show that $\mathbf{P}_3 \subseteq U$. In fact, $x$ and $1 = \frac{1}{3}\cdot 3$ clearly lie in $U$. But then successively,

$$x^2 = \tfrac{1}{2}[(2x^2+1)-1] \quad \text{and} \quad x^3 = (x^2+x^3)-x^2$$

also lie in $U$. Hence $\mathbf{P}_3 \subseteq U$ by Theorem 6.2.2.

**Example 6.2.11**

Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in a vector space $V$. Show that

$$\text{span}\,\{\mathbf{u},\ \mathbf{v}\} = \text{span}\,\{\mathbf{u}+2\mathbf{v},\ \mathbf{u}-\mathbf{v}\}$$

Solution. We have $\text{span}\,\{\mathbf{u}+2\mathbf{v},\ \mathbf{u}-\mathbf{v}\} \subseteq \text{span}\,\{\mathbf{u},\ \mathbf{v}\}$ by Theorem 6.2.2 because both $\mathbf{u}+2\mathbf{v}$ and $\mathbf{u}-\mathbf{v}$ lie in $\text{span}\,\{\mathbf{u},\ \mathbf{v}\}$. On the other hand,

$$\mathbf{u} = \tfrac{1}{3}(\mathbf{u}+2\mathbf{v}) + \tfrac{2}{3}(\mathbf{u}-\mathbf{v}) \quad \text{and} \quad \mathbf{v} = \tfrac{1}{3}(\mathbf{u}+2\mathbf{v}) - \tfrac{1}{3}(\mathbf{u}-\mathbf{v})$$

so $\text{span}\,\{\mathbf{u},\ \mathbf{v}\} \subseteq \text{span}\,\{\mathbf{u}+2\mathbf{v},\ \mathbf{u}-\mathbf{v}\}$, again by Theorem 6.2.2.

# Exercises for 6.2

**Exercise 6.2.1** Which of the following are subspaces of $\mathbf{P}_3$? Support your answer.

a. $U = \{f(x) \mid f(x) \in \mathbf{P}_3,\ f(2) = 1\}$

b. $U = \{xg(x) \mid g(x) \in \mathbf{P}_2\}$

c. $U = \{xg(x) \mid g(x) \in \mathbf{P}_3\}$

d. $U = \{xg(x)+(1-x)h(x) \mid g(x) \text{ and } h(x) \in \mathbf{P}_2\}$

e. $U =$ The set of all polynomials in $\mathbf{P}_3$ with constant term 0

f. $U = \{f(x) \mid f(x) \in \mathbf{P}_3,\ \deg f(x) = 3\}$

**Exercise 6.2.2** Which of the following are subspaces of $\mathbf{M}_{22}$? Support your answer.

a. $U = \left\{ \left[ \begin{array}{cc} a & b \\ 0 & c \end{array} \right] \middle| \ a,\ b, \text{ and } c \text{ in } \mathbb{R} \right\}$

b. $U = \left\{ \left[ \begin{array}{cc} a & b \\ c & d \end{array} \right] \middle| \ a+b = c+d;\ a,\ b,\ c,\ d \text{ in } \mathbb{R} \right\}$

c. $U = \{A \mid A \in \mathbf{M}_{22},\ A = A^T\}$

d. $U = \{A \mid A \in \mathbf{M}_{22},\ AB = 0\}$, $B$ a fixed $2 \times 2$ matrix

e. $U = \{A \mid A \in \mathbf{M}_{22},\ A^2 = A\}$

f. $U = \{A \mid A \in \mathbf{M}_{22},\ A \text{ is not invertible}\}$

g. $U = \{A \mid A \in \mathbf{M}_{22},\ BAC = CAB\}$, $B$ and $C$ fixed $2 \times 2$ matrices

**Exercise 6.2.3** Which of the following are subspaces of $\mathbf{F}[0,\ 1]$? Support your answer.

a. $U = \{f \mid f(0) = 0\}$

b. $U = \{f \mid f(0) = 1\}$

c. $U = \{f \mid f(0) = f(1)\}$

d. $U = \{f \mid f(x) \geq 0$ for all $x$ in $[0,\ 1]\}$

e. $U = \{f \mid f(x) = f(y)$ for all $x$ and $y$ in $[0,\ 1]\}$

f. $U = \{f \mid f(x+y) = f(x) + f(y)$ for all $x$ and $y$ in $[0,\ 1]\}$

g. $U = \{f \mid f$ is integrable and $\int_0^1 f(x)dx = 0\}$

**Exercise 6.2.4** Let $A$ be an $m \times n$ matrix. For which columns $\mathbf{b}$ in $\mathbb{R}^m$ is $U = \{\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n,\ A\mathbf{x} = \mathbf{b}\}$ a subspace of $\mathbb{R}^n$? Support your answer.

**Exercise 6.2.5** Let $\mathbf{x}$ be a vector in $\mathbb{R}^n$ (written as a column), and define $U = \{A\mathbf{x} \mid A \in \mathbf{M}_{mn}\}$.

a. Show that $U$ is a subspace of $\mathbb{R}^m$.

b. Show that $U = \mathbb{R}^m$ if $\mathbf{x} \neq \mathbf{0}$.

**Exercise 6.2.6** Write each of the following as a linear combination of $x+1$, $x^2+x$, and $x^2+2$.

a. $x^2+3x+2$

b. $2x^2-3x+1$

c. $x^2+1$

d. $x$

**Exercise 6.2.7** Determine whether $\mathbf{v}$ lies in span $\{\mathbf{u},\ \mathbf{w}\}$ in each case.

a. $\mathbf{v} = 3x^2-2x-1$; $\mathbf{u} = x^2+1$, $\mathbf{w} = x+2$

b. $\mathbf{v} = x$; $\mathbf{u} = x^2+1$, $\mathbf{w} = x+2$

c. $\mathbf{v} = \left[\begin{array}{rr} 1 & 3 \\ -1 & 1 \end{array}\right]$; $\mathbf{u} = \left[\begin{array}{rr} 1 & -1 \\ 2 & 1 \end{array}\right]$, $\mathbf{w} = \left[\begin{array}{rr} 2 & 1 \\ 1 & 0 \end{array}\right]$

d. $\mathbf{v} = \left[\begin{array}{rr} 1 & -4 \\ 5 & 3 \end{array}\right]$; $\mathbf{u} = \left[\begin{array}{rr} 1 & -1 \\ 2 & 1 \end{array}\right]$, $\mathbf{w} = \left[\begin{array}{rr} 2 & 1 \\ 1 & 0 \end{array}\right]$

**Exercise 6.2.8** Which of the following functions lie in span $\{\cos^2 x,\ \sin^2 x\}$? (Work in $\mathbf{F}[0,\ \pi]$.)

a. $\cos 2x$

b. $1$

c. $x^2$

d. $1+x^2$

**Exercise 6.2.9**

a. Show that $\mathbb{R}^3$ is spanned by $\{(1,\ 0,\ 1),\ (1,\ 1,\ 0),\ (0,\ 1,\ 1)\}$.

b. Show that $\mathbf{P}_2$ is spanned by $\{1+2x^2,\ 3x,\ 1+x\}$.

c. Show that $\mathbf{M}_{22}$ is spanned by $\left\{ \left[\begin{array}{rr} 1 & 0 \\ 0 & 0 \end{array}\right], \left[\begin{array}{rr} 1 & 0 \\ 0 & 1 \end{array}\right], \left[\begin{array}{rr} 0 & 1 \\ 1 & 0 \end{array}\right], \left[\begin{array}{rr} 1 & 1 \\ 0 & 1 \end{array}\right] \right\}$.

**Exercise 6.2.10** If $X$ and $Y$ are two sets of vectors in a vector space $V$, and if $X \subseteq Y$, show that span $X \subseteq$ span $Y$.

**Exercise 6.2.11** Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ denote vectors in a vector space $V$. Show that:

a. span $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\} =$ span $\{\mathbf{u}+\mathbf{v},\ \mathbf{u}+\mathbf{w},\ \mathbf{v}+\mathbf{w}\}$

b. span $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\} =$ span $\{\mathbf{u}-\mathbf{v},\ \mathbf{u}+\mathbf{w},\ \mathbf{w}\}$

**Exercise 6.2.12** Show that

$$\text{span}\,\{\mathbf{v}_1,\ \mathbf{v}_2,\ \dots,\ \mathbf{v}_n,\ \mathbf{0}\} = \text{span}\,\{\mathbf{v}_1,\ \mathbf{v}_2,\ \dots,\ \mathbf{v}_n\}$$

holds for any set of vectors $\{\mathbf{v}_1,\ \mathbf{v}_2,\ \dots,\ \mathbf{v}_n\}$.

**Exercise 6.2.13** If $X$ and $Y$ are nonempty subsets of a vector space $V$ such that span $X =$ span $Y = V$, must there be a vector common to both $X$ and $Y$? Justify your answer.

**Exercise 6.2.14** Is it possible that $\{(1,\ 2,\ 0),\ (1,\ 1,\ 1)\}$ can span the subspace $U = \{(a,\ b,\ 0) \mid a$ and $b$ in $\mathbb{R}\}$?

**Exercise 6.2.15** Describe span $\{\mathbf{0}\}$.

**Exercise 6.2.16** Let $\mathbf{v}$ denote any vector in a vector space $V$. Show that span $\{\mathbf{v}\} =$ span $\{a\mathbf{v}\}$ for any $a \neq 0$.

**Exercise 6.2.17** Determine all subspaces of $\mathbb{R}\mathbf{v}$ where $\mathbf{v} \neq \mathbf{0}$ in some vector space $V$.

**Exercise 6.2.18** Suppose $V =$ span $\{\mathbf{v}_1,\ \mathbf{v}_2,\ \dots,\ \mathbf{v}_n\}$. If $\mathbf{u} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n$ where the $a_i$ are in $\mathbb{R}$ and $a_1 \neq 0$, show that $V =$ span $\{\mathbf{u},\ \mathbf{v}_2,\ \dots,\ \mathbf{v}_n\}$.

**Exercise 6.2.19** If $\mathbf{M}_{nn} =$ span $\{A_1,\ A_2,\ \dots,\ A_k\}$, show that $\mathbf{M}_{nn} =$ span $\{A_1^T,\ A_2^T,\ \dots,\ A_k^T\}$.

**Exercise 6.2.20** If $\mathbf{P}_n = \text{span}\{p_1(x), p_2(x), \dots, p_k(x)\}$ and $a$ is in $\mathbb{R}$, show that $p_i(a) \neq 0$ for some $i$.

**Exercise 6.2.21** Let $U$ be a subspace of a vector space $V$.

a. If $a\mathbf{u}$ is in $U$ where $a \neq 0$, show that $\mathbf{u}$ is in $U$.

b. If $\mathbf{u}$ and $\mathbf{u}+\mathbf{v}$ are in $U$, show that $\mathbf{v}$ is in $U$.

**Exercise 6.2.22** Let $U$ be a nonempty subset of a vector space $V$. Show that $U$ is a subspace of $V$ if and only if $\mathbf{u}_1 + a\mathbf{u}_2$ lies in $U$ for all $\mathbf{u}_1$ and $\mathbf{u}_2$ in $U$ and all $a$ in $\mathbb{R}$.

**Exercise 6.2.23** Let $U = \{p(x) \text{ in } \mathbf{P} \mid p(3) = 0\}$ be the set in Example 6.2.4. Use the factor theorem (see Section 6.5) to show that $U$ consists of multiples of $x-3$; that is, show that $U = \{(x-3)q(x) \mid q(x) \in \mathbf{P}\}$. Use this to show that $U$ is a subspace of $\mathbf{P}$.

**Exercise 6.2.24** Let $A_1, A_2, \dots, A_m$ denote $n \times n$ matrices. If $\mathbf{0} \neq \mathbf{y} \in \mathbb{R}^n$ and $A_1\mathbf{y} = A_2\mathbf{y} = \cdots = A_m\mathbf{y} = \mathbf{0}$, show that $\{A_1, A_2, \dots, A_m\}$ cannot span $\mathbf{M}_{nn}$.

**Exercise 6.2.25** Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be sets of vectors in a vector space, and let

$$X = \left[\begin{array}{c} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_n \end{array}\right] \quad Y = \left[\begin{array}{c} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_n \end{array}\right]$$

as in Exercise 6.1.18.

a. Show that $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subseteq \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ if and only if $AY = X$ for some $n \times n$ matrix $A$.

b. If $X = AY$ where $A$ is invertible, show that $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$.

**Exercise 6.2.26** If $U$ and $W$ are subspaces of a vector space $V$, let $U \cup W = \{\mathbf{v} \mid \mathbf{v} \text{ is in } U \text{ or } \mathbf{v} \text{ is in } W\}$. Show that $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$.

**Exercise 6.2.27** Show that $\mathbf{P}$ cannot be spanned by a finite set of polynomials.

# 6.3 Linear Independence and Dimension

**Definition 6.4 Linear Independence and Dependence**

*As in $\mathbb{R}^n$, a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space $V$ is called* ***linearly independent*** *(or simply* ***independent****) if it satisfies the following condition:*

$$\textit{If} \quad s_1\mathbf{v}_1 + s_2\mathbf{v}_2 + \cdots + s_n\mathbf{v}_n = \mathbf{0}, \qquad \textit{then} \quad s_1 = s_2 = \cdots = s_n = 0.$$

*A set of vectors that is not linearly independent is said to be* ***linearly dependent*** *(or simply* ***dependent****).*

The **trivial linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is the one with every coefficient zero:

$$0\mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_n$$

This is obviously one way of expressing $\mathbf{0}$ as a linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, and they are linearly independent when it is the *only* way.

**Example 6.3.1**

Show that $\{1+x,\ 3x+x^2,\ 2+x-x^2\}$ is independent in $\mathbf{P}_2$.

Solution. Suppose a linear combination of these polynomials vanishes.

$$s_1(1+x)+s_2(3x+x^2)+s_3(2+x-x^2)=0$$

Equating the coefficients of 1, $x$, and $x^2$ gives a set of linear equations.

$$\begin{aligned} s_1 + \quad\;\; + 2s_3 &= 0 \\ s_1 + 3s_2 + \;\; s_3 &= 0 \\ s_2 - \;\; s_3 &= 0 \end{aligned}$$

The only solution is $s_1=s_2=s_3=0$.

**Example 6.3.2**

Show that $\{\sin x,\ \cos x\}$ is independent in the vector space $\mathbf{F}[0,\ 2\pi]$ of functions defined on the interval $[0,\ 2\pi]$.

Solution. Suppose that a linear combination of these functions vanishes.

$$s_1(\sin x)+s_2(\cos x)=0$$

This must hold for *all* values of $x$ in $[0,\ 2\pi]$ (by the definition of equality in $\mathbf{F}[0,\ 2\pi]$). Taking $x=0$ yields $s_2=0$ (because $\sin 0=0$ and $\cos 0=1$). Similarly, $s_1=0$ follows from taking $x=\frac{\pi}{2}$ (because $\sin\frac{\pi}{2}=1$ and $\cos\frac{\pi}{2}=0$).

**Example 6.3.3**

Suppose that $\{\mathbf{u},\ \mathbf{v}\}$ is an independent set in a vector space $V$. Show that $\{\mathbf{u}+2\mathbf{v},\ \mathbf{u}-3\mathbf{v}\}$ is also independent.

Solution. Suppose a linear combination of $\mathbf{u}+2\mathbf{v}$ and $\mathbf{u}-3\mathbf{v}$ vanishes:

$$s(\mathbf{u}+2\mathbf{v})+t(\mathbf{u}-3\mathbf{v})=\mathbf{0}$$

We must deduce that $s=t=0$. Collecting terms involving $\mathbf{u}$ and $\mathbf{v}$ gives

$$(s+t)\mathbf{u}+(2s-3t)\mathbf{v}=\mathbf{0}$$

Because $\{\mathbf{u},\ \mathbf{v}\}$ is independent, this yields linear equations $s+t=0$ and $2s-3t=0$. The only solution is $s=t=0$.

**Example 6.3.4**

Show that any set of polynomials of distinct degrees is independent.

Solution. Let $p_1, p_2, \dots, p_m$ be polynomials where $\deg(p_i) = d_i$. By relabelling if necessary, we may assume that $d_1 > d_2 > \cdots > d_m$. Suppose that a linear combination vanishes:

$$t_1 p_1 + t_2 p_2 + \cdots + t_m p_m = 0$$

where each $t_i$ is in $\mathbb{R}$. As $\deg(p_1) = d_1$, let $ax^{d_1}$ be the term in $p_1$ of highest degree, where $a \neq 0$. Since $d_1 > d_2 > \cdots > d_m$, it follows that $t_1 ax^{d_1}$ is the only term of degree $d_1$ in the linear combination $t_1 p_1 + t_2 p_2 + \cdots + t_m p_m = 0$. This means that $t_1 ax^{d_1} = 0$, whence $t_1 a = 0$, hence $t_1 = 0$ (because $a \neq 0$). But then $t_2 p_2 + \cdots + t_m p_m = 0$ so we can repeat the argument to show that $t_2 = 0$. Continuing, we obtain $t_i = 0$ for each $i$, as desired.

**Example 6.3.5**

Suppose that $A$ is an $n \times n$ matrix such that $A^k = 0$ but $A^{k-1} \neq 0$. Show that $B = \{I, A, A^2, \dots, A^{k-1}\}$ is independent in $\mathbf{M}_{nn}$.

Solution. Suppose $r_0 I + r_1 A + r_2 A^2 + \cdots + r_{k-1} A^{k-1} = 0$. Multiply by $A^{k-1}$:

$$r_0 A^{k-1} + r_1 A^k + r_2 A^{k+1} + \cdots + r_{k-1} A^{2k-2} = 0$$

Since $A^k = 0$, all the higher powers are zero, so this becomes $r_0 A^{k-1} = 0$. But $A^{k-1} \neq 0$, so $r_0 = 0$, and we have $r_1 A^1 + r_2 A^2 + \cdots + r_{k-1} A^{k-1} = 0$. Now multiply by $A^{k-2}$ to conclude that $r_1 = 0$. Continuing, we obtain $r_i = 0$ for each $i$, so $B$ is independent.

The next example collects several useful properties of independence for reference.

**Example 6.3.6**

Let $V$ denote a vector space.

1. If $\mathbf{v} \neq \mathbf{0}$ in $V$, then $\{\mathbf{v}\}$ is an independent set.
2. No independent set of vectors in $V$ can contain the zero vector.

Solution.

1. Let $t\mathbf{v} = \mathbf{0}$, $t$ in $\mathbb{R}$. If $t \neq 0$, then $\mathbf{v} = 1\mathbf{v} = \frac{1}{t}(t\mathbf{v}) = \frac{1}{t}\mathbf{0} = \mathbf{0}$, contrary to assumption. So $t = 0$.
2. If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent and (say) $\mathbf{v}_2 = \mathbf{0}$, then $0\mathbf{v}_1 + 1\mathbf{v}_2 + \cdots + 0\mathbf{v}_k = \mathbf{0}$ is a nontrivial linear combination that vanishes, contrary to the independence of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$.

A set of vectors is independent if $\mathbf{0}$ is a linear combination in a unique way. The following theorem shows that *every* linear combination of these vectors has uniquely determined coefficients, and so extends Theorem 5.2.1.

**Theorem 6.3.1**

*Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly independent set of vectors in a vector space $V$. If a vector $\mathbf{v}$ has two (ostensibly different) representations*

$$\begin{aligned} \mathbf{v} &= s_1\mathbf{v}_1 + s_2\mathbf{v}_2 + \cdots + s_n\mathbf{v}_n \\ \mathbf{v} &= t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_n\mathbf{v}_n \end{aligned}$$

*as linear combinations of these vectors, then $s_1 = t_1, s_2 = t_2, \dots, s_n = t_n$. In other words, every vector in $V$ can be written in a unique way as a linear combination of the $\mathbf{v}_i$.*

Proof. Subtracting the equations given in the theorem gives

$$(s_1 - t_1)\mathbf{v}_1 + (s_2 - t_2)\mathbf{v}_2 + \cdots + (s_n - t_n)\mathbf{v}_n = \mathbf{0}$$

The independence of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ gives $s_i - t_i = 0$ for each $i$, as required. □

The following theorem extends (and proves) Theorem 5.2.4, and is one of the most useful results in linear algebra.

**Theorem 6.3.2: Fundamental Theorem**

*Suppose a vector space $V$ can be spanned by $n$ vectors. If any set of $m$ vectors in $V$ is linearly independent, then $m \leq n$.*

Proof. Let $V = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, and suppose that $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ is an independent set in $V$. Then $\mathbf{u}_1 = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n$ where each $a_i$ is in $\mathbb{R}$. As $\mathbf{u}_1 \neq \mathbf{0}$ (Example 6.3.6), not all of the $a_i$ are zero, say $a_1 \neq 0$ (after relabelling the $\mathbf{v}_i$). Then $V = \operatorname{span}\{\mathbf{u}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ as the reader can verify. Hence, write $\mathbf{u}_2 = b_1\mathbf{u}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \cdots + c_n\mathbf{v}_n$. Then some $c_i \neq 0$ because $\{\mathbf{u}_1, \mathbf{u}_2\}$ is independent; so, as before, $V = \operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$, again after possible relabelling of the $\mathbf{v}_i$. If $m > n$, this procedure continues until all the vectors $\mathbf{v}_i$ are replaced by the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$. In particular, $V = \operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$. But then $\mathbf{u}_{n+1}$ is a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ contrary to the independence of the $\mathbf{u}_i$. Hence, the assumption $m > n$ cannot be valid, so $m \leq n$ and the theorem is proved. □

If $V = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, and if $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ is an independent set in $V$, the above proof shows not only that $m \leq n$ but also that $m$ of the (spanning) vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ can be replaced by the (independent) vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m$ and the resulting set will still span $V$. In this form the result is called the **Steinitz Exchange Lemma**.

**Definition 6.5 Basis of a Vector Space**

*As in $\mathbb{R}^n$, a set $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ of vectors in a vector space $V$ is called a* **basis** *of $V$ if it satisfies the following two conditions:*

1. $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ *is linearly independent*
2. $V = \operatorname{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$

Thus if a set of vectors $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a basis, then *every* vector in $V$ can be written as a linear combination of these vectors in a *unique* way (Theorem 6.3.1). But even more is true: Any two (finite) bases of $V$ contain the same number of vectors.

**Theorem 6.3.3: Invariance Theorem**

*Let $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ and $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ be two bases of a vector space $V$. Then $n = m$.*

Proof. Because $V = \operatorname{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ and $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ is independent, it follows from Theorem 6.3.2 that $m \le n$. Similarly $n \le m$, so $n = m$, as asserted. □

Theorem 6.3.3 guarantees that no matter which basis of $V$ is chosen it contains the same number of vectors as any other basis. Hence there is no ambiguity about the following definition.

**Definition 6.6 Dimension of a Vector Space**

*If $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a basis of the nonzero vector space $V$, the number $n$ of vectors in the basis is called the* **dimension** *of $V$, and we write*

$$\dim V = n$$

*The zero vector space $\{\mathbf{0}\}$ is defined to have dimension $0$:*

$$\dim\{\mathbf{0}\} = 0$$

In our discussion to this point we have always assumed that a basis is nonempty and hence that the dimension of the space is at least 1. However, the zero space $\{\mathbf{0}\}$ has *no* basis (by Example 6.3.6) so our insistence that $\dim\{\mathbf{0}\} = 0$ amounts to saying that the *empty* set of vectors is a basis of $\{\mathbf{0}\}$. Thus the statement that "the dimension of a vector space is the number of vectors in any basis" holds even for the zero space.

We saw in Example 5.2.9 that $\dim(\mathbb{R}^n) = n$ and, if $\mathbf{e}_j$ denotes column $j$ of $I_n$, that $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a basis (called the standard basis). In Example 6.3.7 below, similar considerations apply to the space $\mathbf{M}_{mn}$ of all $m \times n$ matrices; the verifications are left to the reader.

**Example 6.3.7**

The space $\mathbf{M}_{mn}$ has dimension $mn$, and one basis consists of all $m \times n$ matrices with exactly one entry equal to 1 and all other entries equal to 0. We call this the **standard basis** of $\mathbf{M}_{mn}$.

**Example 6.3.8**

Show that $\dim \mathbf{P}_n = n+1$ and that $\{1, x, x^2, \dots, x^n\}$ is a basis, called the **standard basis** of $\mathbf{P}_n$.

Solution. Each polynomial $p(x) = a_0 + a_1x + \cdots + a_nx^n$ in $\mathbf{P}_n$ is clearly a linear combination of $1, x, \dots, x^n$, so $\mathbf{P}_n = \text{span}\{1, x, \dots, x^n\}$. However, if a linear combination of these vectors vanishes, $a_0 1 + a_1x + \cdots + a_nx^n = 0$, then $a_0 = a_1 = \cdots = a_n = 0$ because $x$ is an indeterminate. So $\{1, x, \dots, x^n\}$ is linearly independent and hence is a basis containing $n+1$ vectors. Thus, $\dim(\mathbf{P}_n) = n+1$.

**Example 6.3.9**

If $\mathbf{v} \neq \mathbf{0}$ is any nonzero vector in a vector space $V$, show that $\text{span}\{\mathbf{v}\} = \mathbb{R}\mathbf{v}$ has dimension 1.

Solution. $\{\mathbf{v}\}$ clearly spans $\mathbb{R}\mathbf{v}$, and it is linearly independent by Example 6.3.6. Hence $\{\mathbf{v}\}$ is a basis of $\mathbb{R}\mathbf{v}$, and so $\dim \mathbb{R}\mathbf{v} = 1$.

**Example 6.3.10**

Let $A = \left[\begin{array}{cc} 1 & 1 \\ 0 & 0 \end{array}\right]$ and consider the subspace

$$U = \{X \text{ in } \mathbf{M}_{22} \mid AX = XA\}$$

of $\mathbf{M}_{22}$. Show that $\dim U = 2$ and find a basis of $U$.

Solution. It was shown in Example 6.2.3 that $U$ is a subspace for any choice of the matrix $A$. In the present case, if $X = \left[\begin{array}{cc} x & y \\ z & w \end{array}\right]$ is in $U$, the condition $AX = XA$ gives $z = 0$ and $x = y + w$. Hence each matrix $X$ in $U$ can be written

$$X = \left[\begin{array}{cc} y+w & y \\ 0 & w \end{array}\right] = y\left[\begin{array}{cc} 1 & 1 \\ 0 & 0 \end{array}\right] + w\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]$$

so $U = \text{span}\, B$ where $B = \left\{\left[\begin{array}{cc} 1 & 1 \\ 0 & 0 \end{array}\right], \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]\right\}$. Moreover, the set $B$ is linearly independent (verify this), so it is a basis of $U$ and $\dim U = 2$.

**Example 6.3.11**

Show that the set $V$ of all symmetric $2 \times 2$ matrices is a vector space, and find the dimension of $V$.

Solution. A matrix $A$ is symmetric if $A^T = A$. If $A$ and $B$ lie in $V$, then

$$(A+B)^T = A^T + B^T = A + B \quad \text{and} \quad (kA)^T = kA^T = kA$$

using Theorem 2.1.2. Hence $A+B$ and $kA$ are also symmetric. As the $2 \times 2$ zero matrix is also in

$V$, this shows that $V$ is a vector space (being a subspace of $\mathbf{M}_{22}$). Now a matrix $A$ is symmetric when entries directly across the main diagonal are equal, so each $2 \times 2$ symmetric matrix has the form

$$\left[\begin{array}{cc} a & c \\ c & b \end{array}\right] = a\left[\begin{array}{cc} 1 & 0 \\ 0 & 0 \end{array}\right] + b\left[\begin{array}{cc} 0 & 0 \\ 0 & 1 \end{array}\right] + c\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right]$$

Hence the set $B = \left\{ \left[\begin{array}{cc} 1 & 0 \\ 0 & 0 \end{array}\right], \left[\begin{array}{cc} 0 & 0 \\ 0 & 1 \end{array}\right], \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right] \right\}$ spans $V$, and the reader can verify that $B$ is linearly independent. Thus $B$ is a basis of $V$, so $\dim V = 3$.

It is frequently convenient to alter a basis by multiplying each basis vector by a nonzero scalar. The next example shows that this always produces another basis. The proof is left as Exercise 6.3.22.

**Example 6.3.12**

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be nonzero vectors in a vector space $V$. Given nonzero scalars $a_1, a_2, \dots, a_n$, write $D = \{a_1\mathbf{v}_1, a_2\mathbf{v}_2, \dots, a_n\mathbf{v}_n\}$. If $B$ is independent or spans $V$, the same is true of $D$. In particular, if $B$ is a basis of $V$, so also is $D$.

## Exercises for 6.3

**Exercise 6.3.1** Show that each of the following sets of vectors is independent.

a. $\{1+x, 1-x, x+x^2\}$ in $\mathbf{P}_2$

b. $\{x^2, x+1, 1-x-x^2\}$ in $\mathbf{P}_2$

c. $\left\{ \left[\begin{array}{cc} 1 & 1 \\ 0 & 0 \end{array}\right], \left[\begin{array}{cc} 1 & 0 \\ 1 & 0 \end{array}\right], \left[\begin{array}{cc} 0 & 0 \\ 1 & -1 \end{array}\right], \left[\begin{array}{cc} 0 & 1 \\ 0 & 1 \end{array}\right] \right\}$ in $\mathbf{M}_{22}$

d. $\left\{ \left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right], \left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}\right], \left[\begin{array}{cc} 1 & 0 \\ 1 & 1 \end{array}\right], \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right] \right\}$ in $\mathbf{M}_{22}$

**Exercise 6.3.2** Which of the following subsets of $V$ are independent?

a. $V = \mathbf{P}_2$; $\{x^2+1, x+1, x\}$

b. $V = \mathbf{P}_2$; $\{x^2-x+3, 2x^2+x+5, x^2+5x+1\}$

c. $V = \mathbf{M}_{22}$; $\left\{ \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right], \left[\begin{array}{cc} 1 & 0 \\ 1 & 1 \end{array}\right], \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right] \right\}$

d. $V = \mathbf{M}_{22}$;
$\left\{ \left[\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array}\right], \left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right], \left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right], \left[\begin{array}{cc} 0 & -1 \\ -1 & 0 \end{array}\right] \right\}$

e. $V = \mathbf{F}[1, 2]$; $\left\{\frac{1}{x}, \frac{1}{x^2}, \frac{1}{x^3}\right\}$

f. $V = \mathbf{F}[0, 1]$; $\left\{\frac{1}{x^2+x-6}, \frac{1}{x^2-5x+6}, \frac{1}{x^2-9}\right\}$

**Exercise 6.3.3** Which of the following are independent in $\mathbf{F}[0, 2\pi]$?

a. $\{\sin^2 x, \cos^2 x\}$

b. $\{1, \sin^2 x, \cos^2 x\}$

c. $\{x, \sin^2 x, \cos^2 x\}$

**Exercise 6.3.4** Find all values of $a$ such that the following are independent in $\mathbb{R}^3$.

a. $\{(1, -1, 0), (a, 1, 0), (0, 2, 3)\}$

b. $\{(2,\ a,\ 1),\ (1,\ 0,\ 1),\ (0,\ 1,\ 3)\}$

**Exercise 6.3.5** Show that the following are bases of the space $V$ indicated.

a. $\{(1,\ 1,\ 0),\ (1,\ 0,\ 1),\ (0,\ 1,\ 1)\};\ V = \mathbb{R}^3$

b. $\{(-1,\ 1,\ 1),\ (1,\ -1,\ 1),\ (1,\ 1,\ -1)\};\ V = \mathbb{R}^3$

c. $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\};$ $V = \mathbf{M}_{22}$

d. $\{1+x,\ x+x^2,\ x^2+x^3,\ x^3\};\ V = \mathbf{P}_3$

**Exercise 6.3.6** Exhibit a basis and calculate the dimension of each of the following subspaces of $\mathbf{P}_2$.

a. $\{a(1+x)+b(x+x^2) \mid a \text{ and } b \text{ in } \mathbb{R}\}$

b. $\{a+b(x+x^2) \mid a \text{ and } b \text{ in } \mathbb{R}\}$

c. $\{p(x) \mid p(1) = 0\}$

d. $\{p(x) \mid p(x) = p(-x)\}$

**Exercise 6.3.7** Exhibit a basis and calculate the dimension of each of the following subspaces of $\mathbf{M}_{22}$.

a. $\{A \mid A^T = -A\}$

b. $\left\{ A \,\middle|\, A \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} A \right\}$

c. $\left\{ A \,\middle|\, A \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\}$

d. $\left\{ A \,\middle|\, A \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} A \right\}$

**Exercise 6.3.8** Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and define $U = \{X \mid X \in \mathbf{M}_{22} \text{ and } AX = X\}$.

a. Find a basis of $U$ containing $A$.

b. Find a basis of $U$ not containing $A$.

**Exercise 6.3.9** Show that the set $\mathbb{C}$ of all complex numbers is a vector space with the usual operations, and find its dimension.

**Exercise 6.3.10**

a. Let $V$ denote the set of all $2 \times 2$ matrices with equal column sums. Show that $V$ is a subspace of $\mathbf{M}_{22}$, and compute $\dim V$.

b. Repeat part (a) for $3 \times 3$ matrices.

c. Repeat part (a) for $n \times n$ matrices.

**Exercise 6.3.11**

a. Let $V = \{(x^2+x+1)p(x) \mid p(x) \text{ in } \mathbf{P}_2\}$. Show that $V$ is a subspace of $\mathbf{P}_4$ and find $\dim V$. [*Hint*: If $f(x)g(x) = 0$ in $\mathbf{P}$, then $f(x) = 0$ or $g(x) = 0$.]

b. Repeat with $V = \{(x^2-x)p(x) \mid p(x) \text{ in } \mathbf{P}_3\}$, a subset of $\mathbf{P}_5$.

c. Generalize.

**Exercise 6.3.12** In each case, either prove the assertion or give an example showing that it is false.

a. Every set of four nonzero polynomials in $\mathbf{P}_3$ is a basis.

b. $\mathbf{P}_2$ has a basis of polynomials $f(x)$ such that $f(0) = 0$.

c. $\mathbf{P}_2$ has a basis of polynomials $f(x)$ such that $f(0) = 1$.

d. Every basis of $\mathbf{M}_{22}$ contains a noninvertible matrix.

e. No independent subset of $\mathbf{M}_{22}$ contains a matrix $A$ with $A^2 = 0$.

f. If $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ is independent then, $a\mathbf{u}+b\mathbf{v}+c\mathbf{w} = \mathbf{0}$ for some $a,\ b,\ c$.

g. $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ is independent if $a\mathbf{u}+b\mathbf{v}+c\mathbf{w} = \mathbf{0}$ for some $a,\ b,\ c$.

h. If $\{\mathbf{u},\ \mathbf{v}\}$ is independent, so is $\{\mathbf{u},\ \mathbf{u}+\mathbf{v}\}$.

i. If $\{\mathbf{u},\ \mathbf{v}\}$ is independent, so is $\{\mathbf{u},\ \mathbf{v},\ \mathbf{u}+\mathbf{v}\}$.

j. If $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ is independent, so is $\{\mathbf{u},\ \mathbf{v}\}$.

k. If $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ is independent, so is $\{\mathbf{u}+\mathbf{w},\ \mathbf{v}+\mathbf{w}\}$.

l. If $\{\mathbf{u},\ \mathbf{v},\ \mathbf{w}\}$ is independent, so is $\{\mathbf{u}+\mathbf{v}+\mathbf{w}\}$.

m. If $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$ then $\{\mathbf{u}, \mathbf{v}\}$ is dependent if and only if one is a scalar multiple of the other.

n. If $\dim V = n$, then no set of more than $n$ vectors can be independent.

o. If $\dim V = n$, then no set of fewer than $n$ vectors can span $V$.

**Exercise 6.3.13** Let $A \neq 0$ and $B \neq 0$ be $n \times n$ matrices, and assume that $A$ is symmetric and $B$ is skew-symmetric (that is, $B^T = -B$). Show that $\{A, B\}$ is independent.

**Exercise 6.3.14** Show that every set of vectors containing a dependent set is again dependent.

**Exercise 6.3.15** Show that every nonempty subset of an independent set of vectors is again independent.

**Exercise 6.3.16** Let $f$ and $g$ be functions on $[a, b]$, and assume that $f(a) = 1 = g(b)$ and $f(b) = 0 = g(a)$. Show that $\{f, g\}$ is independent in $\mathbf{F}[a, b]$.

**Exercise 6.3.17** Let $\{A_1, A_2, \dots, A_k\}$ be independent in $\mathbf{M}_{mn}$, and suppose that $U$ and $V$ are invertible matrices of size $m \times m$ and $n \times n$, respectively. Show that $\{UA_1V, UA_2V, \dots, UA_kV\}$ is independent.

**Exercise 6.3.18** Show that $\{\mathbf{v}, \mathbf{w}\}$ is independent if and only if neither $\mathbf{v}$ nor $\mathbf{w}$ is a scalar multiple of the other.

**Exercise 6.3.19** Assume that $\{\mathbf{u}, \mathbf{v}\}$ is independent in a vector space $V$. Write $\mathbf{u}' = a\mathbf{u} + b\mathbf{v}$ and $\mathbf{v}' = c\mathbf{u} + d\mathbf{v}$, where $a$, $b$, $c$, and $d$ are numbers. Show that $\{\mathbf{u}', \mathbf{v}'\}$ is independent if and only if the matrix $\left[ \begin{array}{cc} a & c \\ b & d \end{array} \right]$ is invertible. [*Hint*: Theorem 2.4.5.]

**Exercise 6.3.20** If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent and $\mathbf{w}$ is not in span $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, show that:

a. $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent.

b. $\{\mathbf{v}_1 + \mathbf{w}, \mathbf{v}_2 + \mathbf{w}, \dots, \mathbf{v}_k + \mathbf{w}\}$ is independent.

**Exercise 6.3.21** If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent, show that $\{\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2, \dots, \mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_k\}$ is also independent.

**Exercise 6.3.22** Prove Example 6.3.12.

**Exercise 6.3.23** Let $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{z}\}$ be independent. Which of the following are dependent?

a. $\{\mathbf{u} - \mathbf{v}, \mathbf{v} - \mathbf{w}, \mathbf{w} - \mathbf{u}\}$

b. $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u}\}$

c. $\{\mathbf{u} - \mathbf{v}, \mathbf{v} - \mathbf{w}, \mathbf{w} - \mathbf{z}, \mathbf{z} - \mathbf{u}\}$

d. $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{z}, \mathbf{z} + \mathbf{u}\}$

**Exercise 6.3.24** Let $U$ and $W$ be subspaces of $V$ with bases $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and $\{\mathbf{w}_1, \mathbf{w}_2\}$ respectively. If $U$ and $W$ have only the zero vector in common, show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{w}_1, \mathbf{w}_2\}$ is independent.

**Exercise 6.3.25** Let $\{p, q\}$ be independent polynomials. Show that $\{p, q, pq\}$ is independent if and only if $\deg p \geq 1$ and $\deg q \geq 1$.

**Exercise 6.3.26** If $z$ is a complex number, show that $\{z, z^2\}$ is independent if and only if $z$ is not real.

**Exercise 6.3.27** Let $B = \{A_1, A_2, \dots, A_n\} \subseteq \mathbf{M}_{mn}$, and write $B' = \{A_1^T, A_2^T, \dots, A_n^T\} \subseteq \mathbf{M}_{nm}$. Show that:

a. $B$ is independent if and only if $B'$ is independent.

b. $B$ spans $\mathbf{M}_{mn}$ if and only if $B'$ spans $\mathbf{M}_{nm}$.

**Exercise 6.3.28** If $V = \mathbf{F}[a, b]$ as in Example 6.1.7, show that the set of constant functions is a subspace of dimension 1 ($f$ is **constant** if there is a number $c$ such that $f(x) = c$ for all $x$).

**Exercise 6.3.29**

a. If $U$ is an invertible $n \times n$ matrix and $\{A_1, A_2, \dots, A_{mn}\}$ is a basis of $\mathbf{M}_{mn}$, show that $\{A_1U, A_2U, \dots, A_{mn}U\}$ is also a basis.

b. Show that part (a) fails if $U$ is not invertible. [*Hint*: Theorem 2.4.5.]

**Exercise 6.3.30** Show that $\{(a, b), (a_1, b_1)\}$ is a basis of $\mathbb{R}^2$ if and only if $\{a + bx, a_1 + b_1x\}$ is a basis of $\mathbf{P}_1$.

**Exercise 6.3.31** Find the dimension of the subspace span $\{1, \sin^2\theta, \cos 2\theta\}$ of $\mathbf{F}[0, 2\pi]$.

**Exercise 6.3.32** Show that $\mathbf{F}[0, 1]$ is not finite dimensional.

**Exercise 6.3.33** If $U$ and $W$ are subspaces of $V$, define their intersection $U \cap W$ as follows:

$$U \cap W = \{\mathbf{v} \mid \mathbf{v} \text{ is in both } U \text{ and } W\}$$

a. Show that $U \cap W$ is a subspace contained in $U$ and $W$.

b. Show that $U \cap W = \{\mathbf{0}\}$ if and only if $\{\mathbf{u}, \mathbf{w}\}$ is independent for any nonzero vectors $\mathbf{u}$ in $U$ and $\mathbf{w}$ in $W$.

c. If $B$ and $D$ are bases of $U$ and $W$, and if $U \cap W = \{\mathbf{0}\}$, show that $B \cup D = \{\mathbf{v} \mid \mathbf{v} \text{ is in } B \text{ or } D\}$ is independent.

**Exercise 6.3.34** If $U$ and $W$ are vector spaces, let $V = \{(\mathbf{u}, \mathbf{w}) \mid \mathbf{u} \text{ in } U \text{ and } \mathbf{w} \text{ in } W\}$.

a. Show that $V$ is a vector space if $(\mathbf{u}, \mathbf{w}) + (\mathbf{u}_1, \mathbf{w}_1) = (\mathbf{u} + \mathbf{u}_1, \mathbf{w} + \mathbf{w}_1)$ and $a(\mathbf{u}, \mathbf{w}) = (a\mathbf{u}, a\mathbf{w})$.

b. If $\dim U = m$ and $\dim W = n$, show that $\dim V = m + n$.

c. If $V_1, \dots, V_m$ are vector spaces, let

$$\begin{aligned} V &= V_1 \times \cdots \times V_m \\ &= \{(\mathbf{v}_1, \dots, \mathbf{v}_m) \mid \mathbf{v}_i \in V_i \text{ for each } i\} \end{aligned}$$

denote the space of $n$-tuples from the $V_i$ with componentwise operations (see Exercise 6.1.17). If $\dim V_i = n_i$ for each $i$, show that $\dim V = n_1 + \cdots + n_m$.

**Exercise 6.3.35** Let $\mathbf{D}_n$ denote the set of all functions $f$ from the set $\{1, 2, \dots, n\}$ to $\mathbb{R}$.

a. Show that $\mathbf{D}_n$ is a vector space with pointwise addition and scalar multiplication.

b. Show that $\{S_1, S_2, \dots, S_n\}$ is a basis of $\mathbf{D}_n$ where, for each $k = 1, 2, \dots, n$, the function $S_k$ is defined by $S_k(k) = 1$, whereas $S_k(j) = 0$ if $j \neq k$.

**Exercise 6.3.36** A polynomial $p(x)$ is called **even** if $p(-x) = p(x)$ and **odd** if $p(-x) = -p(x)$. Let $E_n$ and $O_n$ denote the sets of even and odd polynomials in $\mathbf{P}_n$.

a. Show that $E_n$ is a subspace of $\mathbf{P}_n$ and find $\dim E_n$.

b. Show that $O_n$ is a subspace of $\mathbf{P}_n$ and find $\dim O_n$.

**Exercise 6.3.37** Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be independent in a vector space $V$, and let $A$ be an $n \times n$ matrix. Define $\mathbf{u}_1, \dots, \mathbf{u}_n$ by

$$\begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_n \end{bmatrix} = A \begin{bmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_n \end{bmatrix}$$

(See Exercise 6.1.18.) Show that $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is independent if and only if $A$ is invertible.

## 6.4 Finite Dimensional Spaces

Up to this point, we have had no guarantee that an arbitrary vector space *has* a basis—and hence no guarantee that one can speak *at all* of the dimension of $V$. However, Theorem 6.4.1 will show that any space that is spanned by a finite set of vectors has a (finite) basis: The proof requires the following basic lemma, of interest in itself, that gives a way to enlarge a given independent set of vectors.

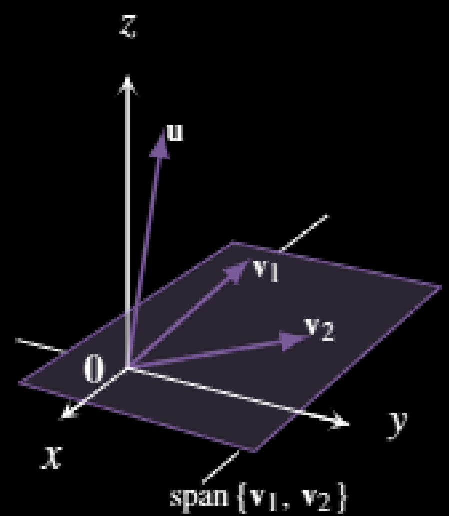
**Lemma 6.4.1: Independent Lemma**

*Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be an independent set of vectors in a vector space $V$. If $\mathbf{u} \in V$ but*[5] *$\mathbf{u} \notin \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, then $\{\mathbf{u}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is also independent.*

Proof. Let $t\mathbf{u} + t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_k\mathbf{v}_k = \mathbf{0}$; we must show that all the coefficients are zero. First, $t = 0$ because, otherwise, $\mathbf{u} = -\frac{t_1}{t}\mathbf{v}_1 - \frac{t_2}{t}\mathbf{v}_2 - \cdots - \frac{t_k}{t}\mathbf{v}_k$ is in $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, contrary to our assumption.

[5]If $X$ is a set, we write $a \in X$ to indicate that $a$ is an element of the set $X$. If $a$ is not an element of $X$, we write $a \notin X$.

Hence $t = 0$. But then $t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_k\mathbf{v}_k = \mathbf{0}$ so the rest of the $t_i$ are zero by the independence of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$. This is what we wanted. □




Note that the converse of Lemma 6.4.1 is also true: if $\{\mathbf{u}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is independent, then $\mathbf{u}$ is not in $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$.

As an illustration, suppose that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is independent in $\mathbb{R}^3$. Then $\mathbf{v}_1$ and $\mathbf{v}_2$ are not parallel, so $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ is a plane through the origin (shaded in the diagram). By Lemma 6.4.1, $\mathbf{u}$ is not in this plane if and only if $\{\mathbf{u}, \mathbf{v}_1, \mathbf{v}_2\}$ is independent.

**Definition 6.7 Finite Dimensional and Infinite Dimensional Vector Spaces**

*A vector space $V$ is called* **finite dimensional** *if it is spanned by a finite set of vectors. Otherwise, $V$ is called* **infinite dimensional***.*

Thus the zero vector space $\{\mathbf{0}\}$ is finite dimensional because $\{\mathbf{0}\}$ is a spanning set.

**Lemma 6.4.2**

*Let $V$ be a finite dimensional vector space. If $U$ is any subspace of $V$, then any independent subset of $U$ can be enlarged to a finite basis of $U$.*

Proof. Suppose that $I$ is an independent subset of $U$. If $\text{span}\, I = U$ then $I$ is already a basis of $U$. If $\text{span}\, I \neq U$, choose $\mathbf{u}_1 \in U$ such that $\mathbf{u}_1 \notin \text{span}\, I$. Hence the set $I \cup \{\mathbf{u}_1\}$ is independent by Lemma 6.4.1. If $\text{span}\,(I \cup \{\mathbf{u}_1\}) = U$ we are done; otherwise choose $\mathbf{u}_2 \in U$ such that $\mathbf{u}_2 \notin \text{span}\,(I \cup \{\mathbf{u}_1\})$. Hence $I \cup \{\mathbf{u}_1, \mathbf{u}_2\}$ is independent, and the process continues. We claim that a basis of $U$ will be reached eventually. Indeed, if no basis of $U$ is ever reached, the process creates arbitrarily large independent sets in $V$. But this is impossible by the fundamental theorem because $V$ is finite dimensional and so is spanned by a finite set of vectors. □

**Theorem 6.4.1**

*Let $V$ be a finite dimensional vector space spanned by $m$ vectors.*

1. *$V$ has a finite basis, and $\dim V \leq m$.*
2. *Every independent set of vectors in $V$ can be enlarged to a basis of $V$ by adding vectors from any fixed basis of $V$.*
3. *If $U$ is a subspace of $V$, then*

   a. *$U$ is finite dimensional and $\dim U \leq \dim V$.*
   
   b. *If $\dim U = \dim V$ then $U = V$.*

**Proof.**

1. If $V = \{\mathbf{0}\}$, then $V$ has an empty basis and $\dim V = 0 \leq m$. Otherwise, let $\mathbf{v} \neq \mathbf{0}$ be a vector in $V$. Then $\{\mathbf{v}\}$ is independent, so (1) follows from Lemma 6.4.2 with $U = V$.

2. We refine the proof of Lemma 6.4.2. Fix a basis $B$ of $V$ and let $I$ be an independent subset of $V$. If $\operatorname{span} I = V$ then $I$ is already a basis of $V$. If $\operatorname{span} I \neq V$, then $B$ is not contained in $I$ (because $B$ spans $V$). Hence choose $\mathbf{b}_1 \in B$ such that $\mathbf{b}_1 \notin \operatorname{span} I$. Hence the set $I \cup \{\mathbf{b}_1\}$ is independent by Lemma 6.4.1. If $\operatorname{span}(I \cup \{\mathbf{b}_1\}) = V$ we are done; otherwise a similar argument shows that $(I \cup \{\mathbf{b}_1, \mathbf{b}_2\})$ is independent for some $\mathbf{b}_2 \in B$. Continue this process. As in the proof of Lemma 6.4.2, a basis of $V$ will be reached eventually.

3. a. This is clear if $U = \{\mathbf{0}\}$. Otherwise, let $\mathbf{u} \neq \mathbf{0}$ in $U$. Then $\{\mathbf{u}\}$ can be enlarged to a finite basis $B$ of $U$ by Lemma 6.4.2, proving that $U$ is finite dimensional. But $B$ is independent in $V$, so $\dim U \leq \dim V$ by the fundamental theorem.

   b. This is clear if $U = \{\mathbf{0}\}$ because $V$ *has* a basis; otherwise, it follows from (2). □

Theorem 6.4.1 shows that a vector space $V$ is finite dimensional if and only if it has a finite basis (possibly empty), and that every subspace of a finite dimensional space is again finite dimensional.

**Example 6.4.1**

Enlarge the independent set $D = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \right\}$ to a basis of $\mathbf{M}_{22}$.

**Solution.** The standard basis of $\mathbf{M}_{22}$ is $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$, so including one of these in $D$ will produce a basis by Theorem 6.4.1. In fact including *any* of these matrices in $D$ produces an independent set (verify), and hence a basis by Theorem 6.4.4. Of course these vectors are not the only possibilities, for example, including $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ works as well.

**Example 6.4.2**

Find a basis of $\mathbf{P}_3$ containing the independent set $\{1+x,\ 1+x^2\}$.

**Solution.** The standard basis of $\mathbf{P}_3$ is $\{1,\ x,\ x^2,\ x^3\}$, so including two of these vectors will do. If we use 1 and $x^3$, the result is $\{1,\ 1+x,\ 1+x^2,\ x^3\}$. This is independent because the polynomials have distinct degrees (Example 6.3.4), and so is a basis by Theorem 6.4.1. Of course, including $\{1,\ x\}$ or $\{1,\ x^2\}$ would *not* work!

**Example 6.4.3**

Show that the space $\mathbf{P}$ of all polynomials is infinite dimensional.

Solution. For each $n \geq 1$, $\mathbf{P}$ has a subspace $\mathbf{P}_n$ of dimension $n+1$. Suppose $\mathbf{P}$ is finite dimensional, say $\dim \mathbf{P} = m$. Then $\dim \mathbf{P}_n \leq \dim \mathbf{P}$ by Theorem 6.4.1, that is $n+1 \leq m$. This is impossible since $n$ is arbitrary, so $\mathbf{P}$ must be infinite dimensional.

The next example illustrates how (2) of Theorem 6.4.1 can be used.

**Example 6.4.4**

If $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_k$ are independent columns in $\mathbb{R}^n$, show that they are the first $k$ columns in some invertible $n \times n$ matrix.

Solution. By Theorem 6.4.1, expand $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_k\}$ to a basis $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_k, \mathbf{c}_{k+1}, \dots, \mathbf{c}_n\}$ of $\mathbb{R}^n$. Then the matrix $A = \left[\begin{array}{cccccc} \mathbf{c}_1 & \mathbf{c}_2 & \dots & \mathbf{c}_k & \mathbf{c}_{k+1} & \dots & \mathbf{c}_n \end{array}\right]$ with this basis as its columns is an $n \times n$ matrix and it is invertible by Theorem 5.2.3.

**Theorem 6.4.2**

*Let $U$ and $W$ be subspaces of the finite dimensional space $V$.*

1. *If $U \subseteq W$, then $\dim U \leq \dim W$.*
2. *If $U \subseteq W$ and $\dim U = \dim W$, then $U = W$.*

Proof. Since $W$ is finite dimensional, (1) follows by taking $V = W$ in part (3) of Theorem 6.4.1. Now assume $\dim U = \dim W = n$, and let $B$ be a basis of $U$. Then $B$ is an independent set in $W$. If $U \neq W$, then $\operatorname{span} B \neq W$, so $B$ can be extended to an independent set of $n+1$ vectors in $W$ by Lemma 6.4.1. This contradicts the fundamental theorem (Theorem 6.3.2) because $W$ is spanned by $\dim W = n$ vectors. Hence $U = W$, proving (2). □

Theorem 6.4.2 is very useful. This was illustrated in Example 5.2.13 for $\mathbb{R}^2$ and $\mathbb{R}^3$; here is another example.

**Example 6.4.5**

If $a$ is a number, let $W$ denote the subspace of all polynomials in $\mathbf{P}_n$ that have $a$ as a root:

$$W = \{p(x) \mid p(x) \in \mathbf{P}_n \text{ and } p(a) = 0\}$$

Show that $\{(x-a), (x-a)^2, \dots, (x-a)^n\}$ is a basis of $W$.

Solution. Observe first that $(x-a), (x-a)^2, \dots, (x-a)^n$ are members of $W$, and that they are independent because they have distinct degrees (Example 6.3.4). Write

$$U = \operatorname{span}\{(x-a), (x-a)^2, \dots, (x-a)^n\}$$

Then we have $U \subseteq W \subseteq \mathbf{P}_n$, $\dim U = n$, and $\dim \mathbf{P}_n = n+1$. Hence $n \leq \dim W \leq n+1$ by Theorem 6.4.2. Since $\dim W$ is an integer, we must have $\dim W = n$ or $\dim W = n+1$. But then $W = U$ or $W = \mathbf{P}_n$, again by Theorem 6.4.2. Because $W \neq \mathbf{P}_n$, it follows that $W = U$, as required.

A set of vectors is called **dependent** if it is *not* independent, that is if some nontrivial linear combination vanishes. The next result is a convenient test for dependence.

**Lemma 6.4.3: Dependent Lemma**

*A set $D = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ of vectors in a vector space $V$ is dependent if and only if some vector in $D$ is a linear combination of the others.*

Proof. Let $\mathbf{v}_2$ (say) be a linear combination of the rest: $\mathbf{v}_2 = s_1\mathbf{v}_1 + s_3\mathbf{v}_3 + \cdots + s_k\mathbf{v}_k$. Then

$$s_1\mathbf{v}_1 + (-1)\mathbf{v}_2 + s_3\mathbf{v}_3 + \cdots + s_k\mathbf{v}_k = \mathbf{0}$$

is a nontrivial linear combination that vanishes, so $D$ is dependent. Conversely, if $D$ is dependent, let $t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_k\mathbf{v}_k = \mathbf{0}$ where some coefficient is nonzero. If (say) $t_2 \neq 0$, then $\mathbf{v}_2 = -\frac{t_1}{t_2}\mathbf{v}_1 - \frac{t_3}{t_2}\mathbf{v}_3 - \cdots - \frac{t_k}{t_2}\mathbf{v}_k$ is a linear combination of the others. □

Lemma 6.4.1 gives a way to enlarge independent sets to a basis; by contrast, Lemma 6.4.3 shows that spanning sets can be cut down to a basis.

**Theorem 6.4.3**

*Let $V$ be a finite dimensional vector space. Any spanning set for $V$ can be cut down (by deleting vectors) to a basis of $V$.*

Proof. Since $V$ is finite dimensional, it has a finite spanning set $S$. Among all spanning sets contained in $S$, choose $S_0$ containing the smallest number of vectors. It suffices to show that $S_0$ is independent (then $S_0$ is a basis, proving the theorem). Suppose, on the contrary, that $S_0$ is not independent. Then, by Lemma 6.4.3, some vector $\mathbf{u} \in S_0$ is a linear combination of the set $S_1 = S_0 \setminus \{\mathbf{u}\}$ of vectors in $S_0$ other than $\mathbf{u}$. It follows that $\text{span } S_0 = \text{span } S_1$, that is, $V = \text{span } S_1$. But $S_1$ has fewer elements than $S_0$ so this contradicts the choice of $S_0$. Hence $S_0$ is independent after all. □

Note that, with Theorem 6.4.1, Theorem 6.4.3 completes the promised proof of Theorem 5.2.6 for the case $V = \mathbb{R}^n$.

**Example 6.4.6**

Find a basis of $\mathbf{P}_3$ in the spanning set $S = \{1,\ x + x^2,\ 2x - 3x^2,\ 1 + 3x - 2x^2,\ x^3\}$.

Solution. Since $\dim \mathbf{P}_3 = 4$, we must eliminate one polynomial from $S$. It cannot be $x^3$ because the span of the rest of $S$ is contained in $\mathbf{P}_2$. But eliminating $1 + 3x - 2x^2$ does leave a basis (verify). Note that $1 + 3x - 2x^2$ is the sum of the first three polynomials in $S$.

Theorems 6.4.1 and 6.4.3 have other useful consequences.

**Theorem 6.4.4**

*Let $V$ be a vector space with $\dim V = n$, and suppose $S$ is a set of exactly $n$ vectors in $V$. Then $S$ is independent if and only if $S$ spans $V$.*

**Proof.** Assume first that $S$ is independent. By Theorem 6.4.1, $S$ is contained in a basis $B$ of $V$. Hence $|S| = n = |B|$ so, since $S \subseteq B$, it follows that $S = B$. In particular $S$ spans $V$.

Conversely, assume that $S$ spans $V$, so $S$ contains a basis $B$ by Theorem 6.4.3. Again $|S| = n = |B|$ so, since $S \supseteq B$, it follows that $S = B$. Hence $S$ is independent. □

One of independence or spanning is often easier to establish than the other when showing that a set of vectors is a basis. For example if $V = \mathbb{R}^n$ it is easy to check whether a subset $S$ of $\mathbb{R}^n$ is orthogonal (hence independent) but checking spanning can be tedious. Here are three more examples.

**Example 6.4.7**

Consider the set $S = \{p_0(x), p_1(x), \dots, p_n(x)\}$ of polynomials in $\mathbf{P}_n$. If $\deg p_k(x) = k$ for each $k$, show that $S$ is a basis of $\mathbf{P}_n$.

**Solution.** The set $S$ is independent—the degrees are distinct—see Example 6.3.4. Hence $S$ is a basis of $\mathbf{P}_n$ by Theorem 6.4.4 because $\dim \mathbf{P}_n = n+1$.

**Example 6.4.8**

Let $V$ denote the space of all symmetric $2 \times 2$ matrices. Find a basis of $V$ consisting of invertible matrices.

**Solution.** We know that $\dim V = 3$ (Example 6.3.11), so what is needed is a set of three invertible, symmetric matrices that (using Theorem 6.4.4) is either independent or spans $V$. The set $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ is independent (verify) and so is a basis of the required type.

**Example 6.4.9**

Let $A$ be any $n \times n$ matrix. Show that there exist $n^2+1$ scalars $a_0, a_1, a_2, \dots, a_{n^2}$ not all zero, such that

$$a_0 I + a_1 A + a_2 A^2 + \cdots + a_{n^2} A^{n^2} = 0$$

where $I$ denotes the $n \times n$ identity matrix.

**Solution.** The space $\mathbf{M}_{nn}$ of all $n \times n$ matrices has dimension $n^2$ by Example 6.3.7. Hence the $n^2+1$ matrices $I, A, A^2, \dots, A^{n^2}$ cannot be independent by Theorem 6.4.4, so a nontrivial linear combination vanishes. This is the desired conclusion.

The result in Example 6.4.9 can be written as $f(A) = 0$ where $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_{n^2} x^{n^2}$. In other words, $A$ satisfies a nonzero polynomial $f(x)$ of degree at most $n^2$. In fact we know that $A$ satisfies a nonzero polynomial of degree $n$ (this is the Cayley-Hamilton theorem—see Theorem 8.7.10), but the brevity of the solution in Example 6.4.6 is an indication of the power of these methods.

If $U$ and $W$ are subspaces of a vector space $V$, there are two related subspaces that are of interest, their **sum** $U+W$ and their **intersection** $U \cap W$, defined by

$$U+W = \{\mathbf{u}+\mathbf{w} \mid \mathbf{u} \in U \text{ and } \mathbf{w} \in W\}$$
$$U \cap W = \{\mathbf{v} \in V \mid \mathbf{v} \in U \text{ and } \mathbf{v} \in W\}$$

It is routine to verify that these are indeed subspaces of $V$, that $U \cap W$ is contained in both $U$ and $W$, and that $U+W$ contains both $U$ and $W$. We conclude this section with a useful fact about the dimensions of these spaces. The proof is a good illustration of how the theorems in this section are used.

**Theorem 6.4.5**

*Suppose that $U$ and $W$ are finite dimensional subspaces of a vector space $V$. Then $U+W$ is finite dimensional and*

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

Proof. Since $U \cap W \subseteq U$, it has a finite basis, say $\{\mathbf{x}_1, \dots, \mathbf{x}_d\}$. Extend it to a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_d, \mathbf{u}_1, \dots, \mathbf{u}_m\}$ of $U$ by Theorem 6.4.1. Similarly extend $\{\mathbf{x}_1, \dots, \mathbf{x}_d\}$ to a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_d, \mathbf{w}_1, \dots, \mathbf{w}_p\}$ of $W$. Then

$$U+W = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_d, \mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_p\}$$

as the reader can verify, so $U+W$ is finite dimensional. For the rest, it suffices to show that $\{\mathbf{x}_1, \dots, \mathbf{x}_d, \mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_p\}$ is independent (verify). Suppose that

$$r_1\mathbf{x}_1 + \cdots + r_d\mathbf{x}_d + s_1\mathbf{u}_1 + \cdots + s_m\mathbf{u}_m + t_1\mathbf{w}_1 + \cdots + t_p\mathbf{w}_p = \mathbf{0} \tag{6.1}$$

where the $r_i$, $s_j$, and $t_k$ are scalars. Then

$$r_1\mathbf{x}_1 + \cdots + r_d\mathbf{x}_d + s_1\mathbf{u}_1 + \cdots + s_m\mathbf{u}_m = -(t_1\mathbf{w}_1 + \cdots + t_p\mathbf{w}_p)$$

is in $U$ (left side) and also in $W$ (right side), and so is in $U \cap W$. Hence $(t_1\mathbf{w}_1 + \cdots + t_p\mathbf{w}_p)$ is a linear combination of $\{\mathbf{x}_1, \dots, \mathbf{x}_d\}$, so $t_1 = \cdots = t_p = 0$, because $\{\mathbf{x}_1, \dots, \mathbf{x}_d, \mathbf{w}_1, \dots, \mathbf{w}_p\}$ is independent. Similarly, $s_1 = \cdots = s_m = 0$, so (6.1) becomes $r_1\mathbf{x}_1 + \cdots + r_d\mathbf{x}_d = \mathbf{0}$. It follows that $r_1 = \cdots = r_d = 0$, as required. □

Theorem 6.4.5 is particularly interesting if $U \cap W = \{\mathbf{0}\}$. Then there are *no* vectors $\mathbf{x}_i$ in the above proof, and the argument shows that if $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_p\}$ are bases of $U$ and $W$ respectively, then $\{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_p\}$ is a basis of $U+W$. In this case $U+W$ is said to be a **direct sum** (written $U \oplus W$); we return to this in Chapter 9.

# Exercises for 6.4

**Exercise 6.4.1** In each case, find a basis for $V$ that includes the vector $\mathbf{v}$.

a. $V = \mathbb{R}^3$, $\mathbf{v} = (1, -1, 1)$

b. $V = \mathbb{R}^3$, $\mathbf{v} = (0, 1, 1)$

c. $V = \mathbf{M}_{22}$, $\mathbf{v} = \left[ \begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array} \right]$

d. $V = \mathbf{P}_2$, $\mathbf{v} = x^2 - x + 1$

**Exercise 6.4.2** In each case, find a basis for $V$ among the given vectors.

a. $V = \mathbb{R}^3$,
$\{(1, 1, -1), (2, 0, 1), (-1, 1, -2), (1, 2, 1)\}$

b. $V = \mathbf{P}_2$, $\{x^2 + 3, x + 2, x^2 - 2x - 1, x^2 + x\}$

**Exercise 6.4.3** In each case, find a basis of $V$ containing $\mathbf{v}$ and $\mathbf{w}$.

a. $V = \mathbb{R}^4$, $\mathbf{v} = (1, -1, 1, -1)$, $\mathbf{w} = (0, 1, 0, 1)$

b. $V = \mathbb{R}^4$, $\mathbf{v} = (0, 0, 1, 1)$, $\mathbf{w} = (1, 1, 1, 1)$

c. $V = \mathbf{M}_{22}$, $\mathbf{v} = \left[ \begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right]$, $\mathbf{w} = \left[ \begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right]$

d. $V = \mathbf{P}_3$, $\mathbf{v} = x^2 + 1$, $\mathbf{w} = x^2 + x$

**Exercise 6.4.4**

a. If $z$ is not a real number, show that $\{z, z^2\}$ is a basis of the real vector space $\mathbb{C}$ of all complex numbers.

b. If $z$ is neither real nor pure imaginary, show that $\{z, \overline{z}\}$ is a basis of $\mathbb{C}$.

**Exercise 6.4.5** In each case use Theorem 6.4.4 to decide if $S$ is a basis of $V$.

a. $V = \mathbf{M}_{22}$;
$S = \left\{ \left[ \begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array} \right], \left[ \begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array} \right], \left[ \begin{array}{cc} 0 & 0 \\ 1 & 1 \end{array} \right], \left[ \begin{array}{cc} 0 & 0 \\ 0 & 1 \end{array} \right] \right\}$

b. $V = \mathbf{P}_3$; $S = \{2x^2, 1 + x, 3, 1 + x + x^2 + x^3\}$

**Exercise 6.4.6**

a. Find a basis of $\mathbf{M}_{22}$ consisting of matrices with the property that $A^2 = A$.

b. Find a basis of $\mathbf{P}_3$ consisting of polynomials whose coefficients sum to 4. What if they sum to 0?

**Exercise 6.4.7** If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a basis of $V$, determine which of the following are bases.

a. $\{\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{w}, \mathbf{v} + \mathbf{w}\}$

b. $\{2\mathbf{u} + \mathbf{v} + 3\mathbf{w}, 3\mathbf{u} + \mathbf{v} - \mathbf{w}, \mathbf{u} - 4\mathbf{w}\}$

c. $\{\mathbf{u}, \mathbf{u} + \mathbf{v} + \mathbf{w}\}$

d. $\{\mathbf{u}, \mathbf{u} + \mathbf{w}, \mathbf{u} - \mathbf{w}, \mathbf{v} + \mathbf{w}\}$

**Exercise 6.4.8**

a. Can two vectors span $\mathbb{R}^3$? Can they be linearly independent? Explain.

b. Can four vectors span $\mathbb{R}^3$? Can they be linearly independent? Explain.

**Exercise 6.4.9** Show that any nonzero vector in a finite dimensional vector space is part of a basis.

**Exercise 6.4.10** If $A$ is a square matrix, show that $\det A = 0$ if and only if some row is a linear combination of the others.

**Exercise 6.4.11** Let $D$, $I$, and $X$ denote finite, nonempty sets of vectors in a vector space $V$. Assume that $D$ is dependent and $I$ is independent. In each case answer yes or no, and defend your answer.

a. If $X \supseteq D$, must $X$ be dependent?

b. If $X \subseteq D$, must $X$ be dependent?

c. If $X \supseteq I$, must $X$ be independent?

d. If $X \subseteq I$, must $X$ be independent?

**Exercise 6.4.12** If $U$ and $W$ are subspaces of $V$ and $\dim U = 2$, show that either $U \subseteq W$ or $\dim (U \cap W) \leq 1$.

**Exercise 6.4.13** Let $A$ be a nonzero $2 \times 2$ matrix and write $U = \{X \text{ in } \mathbf{M}_{22} \mid XA = AX\}$. Show that $\dim U \geq 2$. [*Hint*: $I$ and $A$ are in $U$.]

**Exercise 6.4.14** If $U \subseteq \mathbb{R}^2$ is a subspace, show that $U = \{\mathbf{0}\}$, $U = \mathbb{R}^2$, or $U$ is a line through the origin.

**Exercise 6.4.15** Given $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_k$, and $\mathbf{v}$, let $U = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ and $W = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v}\}$. Show that either $\dim W = \dim U$ or $\dim W = 1 + \dim U$.

**Exercise 6.4.16** Suppose $U$ is a subspace of $\mathbf{P}_1$, $U \neq \{0\}$, and $U \neq \mathbf{P}_1$. Show that either $U = \mathbb{R}$ or $U = \mathbb{R}(a + x)$ for some $a$ in $\mathbb{R}$.

**Exercise 6.4.17** Let $U$ be a subspace of $V$ and assume $\dim V = 4$ and $\dim U = 2$. Does every basis of $V$ result from adding (two) vectors to some basis of $U$? Defend your answer.

**Exercise 6.4.18** Let $U$ and $W$ be subspaces of a vector space $V$.

a. If $\dim V = 3$, $\dim U = \dim W = 2$, and $U \neq W$, show that $\dim (U \cap W) = 1$.

b. Interpret (a.) geometrically if $V = \mathbb{R}^3$.

**Exercise 6.4.19** Let $U \subseteq W$ be subspaces of $V$ with $\dim U = k$ and $\dim W = m$, where $k < m$. If $k < l < m$, show that a subspace $X$ exists where $U \subseteq X \subseteq W$ and $\dim X = l$.

**Exercise 6.4.20** Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a *maximal* independent set in a vector space $V$. That is, no set of more than $n$ vectors $S$ is independent. Show that $B$ is a basis of $V$.

**Exercise 6.4.21** Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a *minimal* spanning set for a vector space $V$. That is, $V$ cannot be spanned by fewer than $n$ vectors. Show that $B$ is a basis of $V$.

**Exercise 6.4.22**

a. Let $p(x)$ and $q(x)$ lie in $\mathbf{P}_1$ and suppose that $p(1) \neq 0$, $q(2) \neq 0$, and $p(2) = 0 = q(1)$. Show that $\{p(x), q(x)\}$ is a basis of $\mathbf{P}_1$. [*Hint*: If $rp(x) + sq(x) = 0$, evaluate at $x = 1$, $x = 2$.]

b. Let $B = \{p_0(x), p_1(x), \dots, p_n(x)\}$ be a set of polynomials in $\mathbf{P}_n$. Assume that there exist numbers $a_0, a_1, \dots, a_n$ such that $p_i(a_i) \neq 0$ for each $i$ but $p_i(a_j) = 0$ if $i$ is different from $j$. Show that $B$ is a basis of $\mathbf{P}_n$.

**Exercise 6.4.23** Let $V$ be the set of all infinite sequences $(a_0, a_1, a_2, \dots)$ of real numbers. Define addition and scalar multiplication by

$$(a_0, a_1, \dots) + (b_0, b_1, \dots) = (a_0 + b_0, a_1 + b_1, \dots)$$

and

$$r(a_0, a_1, \dots) = (ra_0, ra_1, \dots)$$

a. Show that $V$ is a vector space.

b. Show that $V$ is not finite dimensional.

c. [For those with some calculus.] Show that the set of convergent sequences (that is, $\lim\limits_{n \to \infty} a_n$ exists) is a subspace, also of infinite dimension.

**Exercise 6.4.24** Let $A$ be an $n \times n$ matrix of rank $r$. If $U = \{X \text{ in } \mathbf{M}_{nn} \mid AX = 0\}$, show that $\dim U = n(n - r)$. [*Hint*: Exercise 6.3.34.]

**Exercise 6.4.25** Let $U$ and $W$ be subspaces of $V$.

a. Show that $U + W$ is a subspace of $V$ containing both $U$ and $W$.

b. Show that $\operatorname{span}\{\mathbf{u}, \mathbf{w}\} = \mathbb{R}\mathbf{u} + \mathbb{R}\mathbf{w}$ for any vectors $\mathbf{u}$ and $\mathbf{w}$.

c. Show that

$$\begin{aligned} &\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_n\} \\ &= \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_m\} + \operatorname{span}\{\mathbf{w}_1, \dots, \mathbf{w}_n\} \end{aligned}$$

for any vectors $\mathbf{u}_i$ in $U$ and $\mathbf{w}_j$ in $W$.

**Exercise 6.4.26** If $A$ and $B$ are $m \times n$ matrices, show that $\operatorname{rank}(A + B) \leq \operatorname{rank} A + \operatorname{rank} B$. [*Hint*: If $U$ and $V$ are the column spaces of $A$ and $B$, respectively, show that the column space of $A + B$ is contained in $U + V$ and that $\dim (U + V) \leq \dim U + \dim V$. (See Theorem 6.4.5.)]

# Exercises for 7.1

**Exercise 7.1.1** Show that each of the following functions is a linear transformation.

a. $T : \mathbb{R}^2 \to \mathbb{R}^2$; $T(x,\ y) = (x,\ -y)$ (reflection in the $x$ axis)

b. $T : \mathbb{R}^3 \to \mathbb{R}^3$; $T(x,\ y,\ z) = (x,\ y,\ -z)$ (reflection in the $x$-$y$ plane)

c. $T : \mathbb{C} \to \mathbb{C}$; $T(z) = \overline{z}$ (conjugation)

d. $T : \mathbf{M}_{mn} \to \mathbf{M}_{kl}$; $T(A) = PAQ$, $P$ a $k \times m$ matrix, $Q$ an $n \times l$ matrix, both fixed

e. $T : \mathbf{M}_{nn} \to \mathbf{M}_{nn}$; $T(A) = A^T + A$

f. $T : \mathbf{P}_n \to \mathbb{R}$; $T\,[p(x)] = p(0)$

g. $T : \mathbf{P}_n \to \mathbb{R}$; $T(r_0 + r_1x + \cdots + r_nx^n) = r_n$

h. $T : \mathbb{R}^n \to \mathbb{R}$; $T(\mathbf{x}) = \mathbf{x} \cdot \mathbf{z}$, $\mathbf{z}$ a fixed vector in $\mathbb{R}^n$

i. $T : \mathbf{P}_n \to \mathbf{P}_n$; $T\,[p(x)] = p(x+1)$

j. $T : \mathbb{R}^n \to V$; $T(r_1,\ \cdots,\ r_n) = r_1\mathbf{e}_1 + \cdots + r_n\mathbf{e}_n$ where $\{\mathbf{e}_1,\ \ldots,\ \mathbf{e}_n\}$ is a fixed basis of $V$

k. $T : V \to \mathbb{R}$; $T(r_1\mathbf{e}_1 + \cdots + r_n\mathbf{e}_n) = r_1$, where $\{\mathbf{e}_1,\ \ldots,\ \mathbf{e}_n\}$ is a fixed basis of $V$

**Exercise 7.1.2** In each case, show that $T$ is *not* a linear transformation.

a. $T : \mathbf{M}_{nn} \to \mathbb{R}$; $T(A) = \det A$

b. $T : \mathbf{M}_{nm} \to \mathbb{R}$; $T(A) = \operatorname{rank} A$

c. $T : \mathbb{R} \to \mathbb{R}$; $T(x) = x^2$

d. $T : V \to V$; $T(\mathbf{v}) = \mathbf{v} + \mathbf{u}$ where $\mathbf{u} \neq \mathbf{0}$ is a fixed vector in $V$ ($T$ is called the **translation** by $\mathbf{u}$)

**Exercise 7.1.3** In each case, assume that $T$ is a linear transformation.

a. If $T : V \to \mathbb{R}$ and $T(\mathbf{v}_1) = 1$, $T(\mathbf{v}_2) = -1$, find $T(3\mathbf{v}_1 - 5\mathbf{v}_2)$.

b. If $T : V \to \mathbb{R}$ and $T(\mathbf{v}_1) = 2$, $T(\mathbf{v}_2) = -3$, find $T(3\mathbf{v}_1 + 2\mathbf{v}_2)$.

c. If $T : \mathbb{R}^2 \to \mathbb{R}^2$ and $T\left[\begin{array}{r} 1 \\ 3 \end{array}\right] = \left[\begin{array}{r} 1 \\ 1 \end{array}\right]$, $T\left[\begin{array}{r} 1 \\ 1 \end{array}\right] = \left[\begin{array}{r} 0 \\ 1 \end{array}\right]$, find $T\left[\begin{array}{r} -1 \\ 3 \end{array}\right]$.

d. If $T : \mathbb{R}^2 \to \mathbb{R}^2$ and $T\left[\begin{array}{r} 1 \\ -1 \end{array}\right] = \left[\begin{array}{r} 0 \\ 1 \end{array}\right]$, $T\left[\begin{array}{r} 1 \\ 1 \end{array}\right] = \left[\begin{array}{r} 1 \\ 0 \end{array}\right]$, find $T\left[\begin{array}{r} 1 \\ -7 \end{array}\right]$.

e. If $T : \mathbf{P}_2 \to \mathbf{P}_2$ and $T(x+1) = x$, $T(x-1) = 1$, $T(x^2) = 0$, find $T(2 + 3x - x^2)$.

f. If $T : \mathbf{P}_2 \to \mathbb{R}$ and $T(x+2) = 1$, $T(1) = 5$, $T(x^2 + x) = 0$, find $T(2 - x + 3x^2)$.

**Exercise 7.1.4** In each case, find a linear transformation with the given properties and compute $T(\mathbf{v})$.

a. $T : \mathbb{R}^2 \to \mathbb{R}^3$; $T(1,\ 2) = (1,\ 0,\ 1)$, $T(-1,\ 0) = (0,\ 1,\ 1)$; $\mathbf{v} = (2,\ 1)$

b. $T : \mathbb{R}^2 \to \mathbb{R}^3$; $T(2,\ -1) = (1,\ -1,\ 1)$, $T(1,\ 1) = (0,\ 1,\ 0)$; $\mathbf{v} = (-1,\ 2)$

c. $T : \mathbf{P}_2 \to \mathbf{P}_3$; $T(x^2) = x^3$, $T(x+1) = 0$, $T(x-1) = x$; $\mathbf{v} = x^2 + x + 1$

d. $T : \mathbf{M}_{22} \to \mathbb{R}$; $T\left[\begin{array}{rr} 1 & 0 \\ 0 & 0 \end{array}\right] = 3$, $T\left[\begin{array}{rr} 0 & 1 \\ 1 & 0 \end{array}\right] = -1$, $T\left[\begin{array}{rr} 1 & 0 \\ 1 & 0 \end{array}\right] = 0 = T\left[\begin{array}{rr} 0 & 0 \\ 0 & 1 \end{array}\right]$; $\mathbf{v} = \left[\begin{array}{rr} a & b \\ c & d \end{array}\right]$

**Exercise 7.1.5** If $T : V \to V$ is a linear transformation, find $T(\mathbf{v})$ and $T(\mathbf{w})$ if:

a. $T(\mathbf{v} + \mathbf{w}) = \mathbf{v} - 2\mathbf{w}$ and $T(2\mathbf{v} - \mathbf{w}) = 2\mathbf{v}$

b. $T(\mathbf{v} + 2\mathbf{w}) = 3\mathbf{v} - \mathbf{w}$ and $T(\mathbf{v} - \mathbf{w}) = 2\mathbf{v} - 4\mathbf{w}$

**Exercise 7.1.6** If $T : V \to W$ is a linear transformation, show that $T(\mathbf{v} - \mathbf{v}_1) = T(\mathbf{v}) - T(\mathbf{v}_1)$ for all $\mathbf{v}$ and $\mathbf{v}_1$ in $V$.

**Exercise 7.1.7** Let $\{\mathbf{e}_1,\ \mathbf{e}_2\}$ be the standard basis of $\mathbb{R}^2$. Is it possible to have a linear transformation $T$ such that $T(\mathbf{e}_1)$ lies in $\mathbb{R}$ while $T(\mathbf{e}_2)$ lies in $\mathbb{R}^2$? Explain your answer.

**Exercise 7.1.8** Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis of $V$ and let $T : V \to V$ be a linear transformation.

a. If $T(\mathbf{v}_i) = \mathbf{v}_i$ for each $i$, show that $T = 1_V$.

b. If $T(\mathbf{v}_i) = -\mathbf{v}_i$ for each $i$, show that $T = -1$ is the scalar operator (see Example 7.1.1).

**Exercise 7.1.9** If $A$ is an $m \times n$ matrix, let $C_k(A)$ denote column $k$ of $A$. Show that $C_k : \mathbf{M}_{mn} \to \mathbb{R}^m$ is a linear transformation for each $k = 1, \dots, n$.

**Exercise 7.1.10** Let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be a basis of $\mathbb{R}^n$. Given $k$, $1 \leq k \leq n$, define $P_k : \mathbb{R}^n \to \mathbb{R}^n$ by $P_k(r_1\mathbf{e}_1 + \cdots + r_n\mathbf{e}_n) = r_k\mathbf{e}_k$. Show that $P_k$ a linear transformation for each $k$.

**Exercise 7.1.11** Let $S : V \to W$ and $T : V \to W$ be linear transformations. Given $a$ in $\mathbb{R}$, define functions $(S+T) : V \to W$ and $(aT) : V \to W$ by $(S+T)(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v})$ and $(aT)(\mathbf{v}) = aT(\mathbf{v})$ for all $\mathbf{v}$ in $V$. Show that $S+T$ and $aT$ are linear transformations.

**Exercise 7.1.12** Describe all linear transformations $T : \mathbb{R} \to V$.

**Exercise 7.1.13** Let $V$ and $W$ be vector spaces, let $V$ be finite dimensional, and let $\mathbf{v} \neq \mathbf{0}$ in $V$. Given any $\mathbf{w}$ in $W$, show that there exists a linear transformation $T : V \to W$ with $T(\mathbf{v}) = \mathbf{w}$. [*Hint*: Theorem 6.4.1 and Theorem 7.1.3.]

**Exercise 7.1.14** Given $\mathbf{y}$ in $\mathbb{R}^n$, define $S_\mathbf{y} : \mathbb{R}^n \to \mathbb{R}$ by $S_\mathbf{y}(\mathbf{x}) = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}$ in $\mathbb{R}^n$ (where $\cdot$ is the dot product introduced in Section 5.3).

a. Show that $S_\mathbf{y} : \mathbb{R}^n \to \mathbb{R}$ is a linear transformation for any $\mathbf{y}$ in $\mathbb{R}^n$.

b. Show that every linear transformation $T : \mathbb{R}^n \to \mathbb{R}$ arises in this way; that is, $T = S_\mathbf{y}$ for some $\mathbf{y}$ in $\mathbb{R}^n$. [*Hint*: If $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$, write $S_\mathbf{y}(\mathbf{e}_i) = y_i$ for each $i$. Use Theorem 7.1.1.]

**Exercise 7.1.15** Let $T : V \to W$ be a linear transformation.

a. If $U$ is a subspace of $V$, show that $T(U) = \{T(\mathbf{u}) \mid \mathbf{u} \text{ in } U\}$ is a subspace of $W$ (called the **image** of $U$ under $T$).

b. If $P$ is a subspace of $W$, show that $\{\mathbf{v} \text{ in } V \mid T(\mathbf{v}) \text{ in } P\}$ is a subspace of $V$ (called the **preimage** of $P$ under $T$).

**Exercise 7.1.16** Show that differentiation is the only linear transformation $\mathbf{P}_n \to \mathbf{P}_n$ that satisfies $T(x^k) = kx^{k-1}$ for each $k = 0, 1, 2, \dots, n$.

**Exercise 7.1.17** Let $T : V \to W$ be a linear transformation and let $\mathbf{v}_1, \dots, \mathbf{v}_n$ denote vectors in $V$.

a. If $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is linearly independent, show that $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is also independent.

b. Find $T : \mathbb{R}^2 \to \mathbb{R}^2$ for which the converse of part (a) is false.

**Exercise 7.1.18** Suppose $T : V \to V$ is a linear operator with the property that $T[T(\mathbf{v})] = \mathbf{v}$ for all $\mathbf{v}$ in $V$. (For example, transposition in $\mathbf{M}_{nn}$ or conjugation in $\mathbb{C}$.) If $\mathbf{v} \neq \mathbf{0}$ in $V$, show that $\{\mathbf{v}, T(\mathbf{v})\}$ is linearly independent if and only if $T(\mathbf{v}) \neq \mathbf{v}$ and $T(\mathbf{v}) \neq -\mathbf{v}$.

**Exercise 7.1.19** If $a$ and $b$ are real numbers, define $T_{a,\,b} : \mathbb{C} \to \mathbb{C}$ by $T_{a,\,b}(r+si) = ra + sbi$ for all $r+si$ in $\mathbb{C}$.

a. Show that $T_{a,\,b}$ is linear and $T_{a,\,b}(\overline{z}) = \overline{T_{a,\,b}(z)}$ for all $z$ in $\mathbb{C}$. (Here $\overline{z}$ denotes the conjugate of $z$.)

b. If $T : \mathbb{C} \to \mathbb{C}$ is linear and $T(\overline{z}) = \overline{T(z)}$ for all $z$ in $\mathbb{C}$, show that $T = T_{a,\,b}$ for some real $a$ and $b$.

**Exercise 7.1.20** Show that the following conditions are equivalent for a linear transformation $T : \mathbf{M}_{22} \to \mathbf{M}_{22}$.

1. $\operatorname{tr}[T(A)] = \operatorname{tr} A$ for all $A$ in $\mathbf{M}_{22}$.

2. $T \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} = r_{11}B_{11} + r_{12}B_{12} + r_{21}B_{21} + r_{22}B_{22}$ for matrices $B_{ij}$ such that $\operatorname{tr} B_{11} = 1 = \operatorname{tr} B_{22}$ and $\operatorname{tr} B_{12} = 0 = \operatorname{tr} B_{21}$.

**Exercise 7.1.21** Given $a$ in $\mathbb{R}$, consider the **evaluation** map $E_a : \mathbf{P}_n \to \mathbb{R}$ defined in Example 7.1.3.

a. Show that $E_a$ is a linear transformation satisfying the additional condition that $E_a(x^k) = [E_a(x)]^k$ holds for all $k = 0, 1, 2, \dots$. [*Note*: $x^0 = 1$.]

b. If $T : \mathbf{P}_n \to \mathbb{R}$ is a linear transformation satisfying $T(x^k) = [T(x)]^k$ for all $k = 0, 1, 2, \dots$, show that $T = E_a$ for some $a$ in $R$.

**Exercise 7.1.22** If $T : \mathbf{M}_{nn} \to \mathbb{R}$ is any linear transformation satisfying $T(AB) = T(BA)$ for all $A$ and $B$ in $\mathbf{M}_{nn}$, show that there exists a number $k$ such that $T(A) = k \operatorname{tr} A$ for all $A$. (See Lemma 5.5.1.) [*Hint*: Let $E_{ij}$ denote the $n \times n$ matrix with 1 in the $(i,\ j)$ position and zeros elsewhere.

Show that $E_{ik}E_{lj} = \begin{cases} 0 & \text{if } k \neq l \\ E_{ij} & \text{if } k = l \end{cases}$. Use this to show that $T(E_{ij}) = 0$ if $i \neq j$ and $T(E_{11}) = T(E_{22}) = \cdots = T(E_{nn})$. Put $k = T(E_{11})$ and use the fact that $\{E_{ij} \mid 1 \leq i,\ j \leq n\}$ is a basis of $\mathbf{M}_{nn}$.]

**Exercise 7.1.23** Let $T : \mathbb{C} \to \mathbb{C}$ be a linear transformation of the real vector space $\mathbb{C}$ and assume that $T(a) = a$ for every real number $a$. Show that the following are equivalent:

a. $T(zw) = T(z)T(w)$ for all $z$ and $w$ in $\mathbb{C}$.

b. Either $T = 1_{\mathbb{C}}$ or $T(z) = \overline{z}$ for each $z$ in $\mathbb{C}$ (where $\overline{z}$ denotes the conjugate).

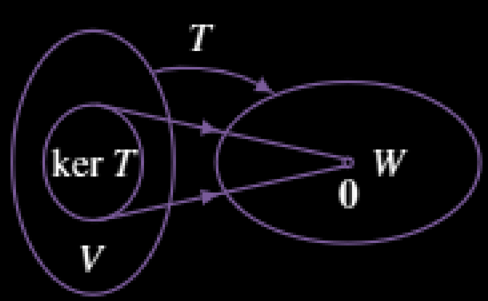
# 7.2 Kernel and Image of a Linear Transformation

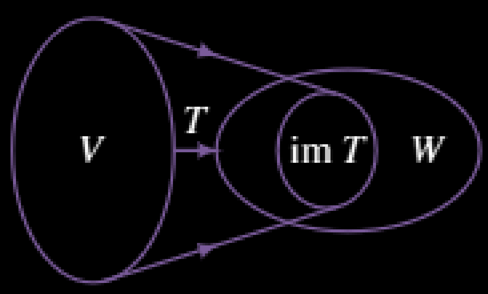
This section is devoted to two important subspaces associated with a linear transformation $T : V \to W$.

**Definition 7.2 Kernel and Image of a Linear Transformation**

*The* **kernel** *of* $T$ *(denoted* $\ker T$*) and the* **image** *of* $T$ *(denoted* $\operatorname{im} T$ *or* $T(V)$*) are defined by*

$$\ker T = \{\mathbf{v} \text{ in } V \mid T(\mathbf{v}) = \mathbf{0}\}$$
$$\operatorname{im} T = \{T(\mathbf{v}) \mid \mathbf{v} \text{ in } V\} = T(V)$$




The kernel of $T$ is often called the **nullspace** of $T$ because it consists of all vectors $\mathbf{v}$ in $V$ satisfying the *condition* that $T(\mathbf{v}) = \mathbf{0}$. The image of $T$ is often called the **range** of $T$ and consists of all vectors $\mathbf{w}$ in $W$ of the *form* $\mathbf{w} = T(\mathbf{v})$ for some $\mathbf{v}$ in $V$. These subspaces are depicted in the diagrams.

**Example 7.2.1**

Let $T_A : \mathbb{R}^n \to \mathbb{R}^m$ be the linear transformation induced by the $m \times n$ matrix $A$, that is $T_A(\mathbf{x}) = A\mathbf{x}$ for all columns $\mathbf{x}$ in $\mathbb{R}^n$. Then

$$\ker T_A = \{\mathbf{x} \mid A\mathbf{x} = \mathbf{0}\} = \operatorname{null} A \quad \text{and}$$
$$\operatorname{im} T_A = \{A\mathbf{x} \mid \mathbf{x} \text{ in } \mathbb{R}^n\} = \operatorname{im} A$$

Hence the following theorem extends Example 5.1.2.

The dimension theorem and Example 7.2.2 give

$$\begin{aligned}\text{rank } A &= \text{rank } T_A = \dim(\text{im } T_A) = n - \dim(\ker T_A)\\ \text{rank } B &= \text{rank } T_B = \dim(\text{im } T_B) = n - \dim(\ker T_B)\end{aligned}$$

so it suffices to show that $\ker T_A = \ker T_B$. Now $A\mathbf{x} = \mathbf{0}$ implies that $B\mathbf{x} = A^T A\mathbf{x} = \mathbf{0}$, so $\ker T_A$ is contained in $\ker T_B$. On the other hand, if $B\mathbf{x} = \mathbf{0}$, then $A^T A\mathbf{x} = \mathbf{0}$, so

$$\|A\mathbf{x}\|^2 = (A\mathbf{x})^T(A\mathbf{x}) = \mathbf{x}^T A^T A\mathbf{x} = \mathbf{x}^T\mathbf{0} = 0$$

This implies that $A\mathbf{x} = \mathbf{0}$, so $\ker T_B$ is contained in $\ker T_A$.

# Exercises for 7.2

**Exercise 7.2.1** For each matrix $A$, find a basis for the kernel and image of $T_A$, and find the rank and nullity of $T_A$.

a. $\left[\begin{array}{rrrr} 1 & 2 & -1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & -3 & 2 & 0 \end{array}\right]$

b. $\left[\begin{array}{rrrr} 2 & 1 & -1 & 3 \\ 1 & 0 & 3 & 1 \\ 1 & 1 & -4 & 2 \end{array}\right]$

c. $\left[\begin{array}{rrr} 1 & 2 & -1 \\ 3 & 1 & 2 \\ 4 & -1 & 5 \\ 0 & 2 & -2 \end{array}\right]$

d. $\left[\begin{array}{rrr} 2 & 1 & 0 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \\ 0 & 3 & -6 \end{array}\right]$

**Exercise 7.2.2** In each case, (i) find a basis of $\ker T$, and (ii) find a basis of $\text{im } T$. You may assume that $T$ is linear.

a. $T : \mathbf{P}_2 \to \mathbb{R}^2$; $T(a + bx + cx^2) = (a, b)$

b. $T : \mathbf{P}_2 \to \mathbb{R}^2$; $T(p(x)) = (p(0), p(1))$

c. $T : \mathbb{R}^3 \to \mathbb{R}^3$; $T(x, y, z) = (x + y, x + y, 0)$

d. $T : \mathbb{R}^3 \to \mathbb{R}^4$; $T(x, y, z) = (x, x, y, y)$

e. $T : \mathbf{M}_{22} \to \mathbf{M}_{22}$; $T\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] = \left[\begin{array}{cc} a+b & b+c \\ c+d & d+a \end{array}\right]$

f. $T : \mathbf{M}_{22} \to \mathbb{R}$; $T\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] = a + d$

g. $T : \mathbf{P}_n \to \mathbb{R}$; $T(r_0 + r_1x + \cdots + r_nx^n) = r_n$

h. $T : \mathbb{R}^n \to \mathbb{R}$; $T(r_1, r_2, \dots, r_n) = r_1 + r_2 + \cdots + r_n$

i. $T : \mathbf{M}_{22} \to \mathbf{M}_{22}$; $T(X) = XA - AX$, where $A = \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right]$

j. $T : \mathbf{M}_{22} \to \mathbf{M}_{22}$; $T(X) = XA$, where $A = \left[\begin{array}{cc} 1 & 1 \\ 0 & 0 \end{array}\right]$

**Exercise 7.2.3** Let $P : V \to \mathbb{R}$ and $Q : V \to \mathbb{R}$ be linear transformations, where $V$ is a vector space. Define $T : V \to \mathbb{R}^2$ by $T(\mathbf{v}) = (P(\mathbf{v}), Q(\mathbf{v}))$.

a. Show that $T$ is a linear transformation.

b. Show that $\ker T = \ker P \cap \ker Q$, the set of vectors in both $\ker P$ and $\ker Q$.

**Exercise 7.2.4** In each case, find a basis $B = \{\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ of $V$ such that $\{\mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ is a basis of $\ker T$, and verify Theorem 7.2.5.

a. $T : \mathbb{R}^3 \to \mathbb{R}^4$; $T(x, y, z) = (x - y + 2z, x + y - z, 2x + z, 2y - 3z)$

b. $T : \mathbb{R}^3 \to \mathbb{R}^4$; $T(x, y, z) = (x + y + z, 2x - y + 3z, z - 3y, 3x + 4z)$

**Exercise 7.2.5** Show that every matrix $X$ in $\mathbf{M}_{nn}$ has the form $X = A^T - 2A$ for some matrix $A$ in $\mathbf{M}_{nn}$. [*Hint*: The dimension theorem.]

**Exercise 7.2.6** In each case either prove the statement or give an example in which it is false. Throughout, let $T : V \to W$ be a linear transformation where $V$ and $W$ are finite dimensional.

a. If $V = W$, then $\ker T \subseteq \operatorname{im} T$.

b. If $\dim V = 5$, $\dim W = 3$, and $\dim(\ker T) = 2$, then $T$ is onto.

c. If $\dim V = 5$ and $\dim W = 4$, then $\ker T \neq \{\mathbf{0}\}$.

d. If $\ker T = V$, then $W = \{\mathbf{0}\}$.

e. If $W = \{\mathbf{0}\}$, then $\ker T = V$.

f. If $W = V$, and $\operatorname{im} T \subseteq \ker T$, then $T = 0$.

g. If $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is a basis of $V$ and $T(\mathbf{e}_1) = \mathbf{0} = T(\mathbf{e}_2)$, then $\dim(\operatorname{im} T) \leq 1$.

h. If $\dim(\ker T) \leq \dim W$, then $\dim W \geq \frac{1}{2} \dim V$.

i. If $T$ is one-to-one, then $\dim V \leq \dim W$.

j. If $\dim V \leq \dim W$, then $T$ is one-to-one.

k. If $T$ is onto, then $\dim V \geq \dim W$.

l. If $\dim V \geq \dim W$, then $T$ is onto.

m. If $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_k)\}$ is independent, then $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is independent.

n. If $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ spans $V$, then $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_k)\}$ spans $W$.

**Exercise 7.2.7** Show that linear independence is preserved by one-to-one transformations and that spanning sets are preserved by onto transformations. More precisely, if $T : V \to W$ is a linear transformation, show that:

a. If $T$ is one-to-one and $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is independent in $V$, then $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is independent in $W$.

b. If $T$ is onto and $V = \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, then $W = \operatorname{span}\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$.

**Exercise 7.2.8** Given $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in a vector space $V$, define $T : \mathbb{R}^n \to V$ by $T(r_1, \dots, r_n) = r_1\mathbf{v}_1 + \cdots + r_n\mathbf{v}_n$. Show that $T$ is linear, and that:

a. $T$ is one-to-one if and only if $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is independent.

b. $T$ is onto if and only if $V = \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

**Exercise 7.2.9** Let $T : V \to V$ be a linear transformation where $V$ is finite dimensional. Show that exactly one of (i) and (ii) holds: (i) $T(\mathbf{v}) = \mathbf{0}$ for some $\mathbf{v} \neq \mathbf{0}$ in $V$; (ii) $T(\mathbf{x}) = \mathbf{v}$ has a solution $\mathbf{x}$ in $V$ for every $\mathbf{v}$ in $V$.

**Exercise 7.2.10** Let $T : \mathbf{M}_{nn} \to \mathbb{R}$ denote the trace map: $T(A) = \operatorname{tr} A$ for all $A$ in $\mathbf{M}_{nn}$. Show that $\dim(\ker T) = n^2 - 1$.

**Exercise 7.2.11** Show that the following are equivalent for a linear transformation $T : V \to W$.

1. $\ker T = V$
2. $\operatorname{im} T = \{\mathbf{0}\}$
3. $T = 0$

**Exercise 7.2.12** Let $A$ and $B$ be $m \times n$ and $k \times n$ matrices, respectively. Assume that $A\mathbf{x} = \mathbf{0}$ implies $B\mathbf{x} = \mathbf{0}$ for every $n$-column $\mathbf{x}$. Show that $\operatorname{rank} A \geq \operatorname{rank} B$.
[*Hint*: Theorem 7.2.4.]

**Exercise 7.2.13** Let $A$ be an $m \times n$ matrix of rank $r$. Thinking of $\mathbb{R}^n$ as rows, define $V = \{\mathbf{x} \text{ in } \mathbb{R}^m \mid \mathbf{x}A = \mathbf{0}\}$. Show that $\dim V = m - r$.

**Exercise 7.2.14** Consider

$$V = \left\{ \left[ \begin{array}{cc} a & b \\ c & d \end{array} \right] \middle| a + c = b + d \right\}$$

a. Consider $S : \mathbf{M}_{22} \to \mathbb{R}$ with $S\left[ \begin{array}{cc} a & b \\ c & d \end{array} \right] = a + c - b - d$. Show that $S$ is linear and onto and that $V$ is a subspace of $\mathbf{M}_{22}$. Compute $\dim V$.

b. Consider $T : V \to \mathbb{R}$ with $T\left[ \begin{array}{cc} a & b \\ c & d \end{array} \right] = a + c$. Show that $T$ is linear and onto, and use this information to compute $\dim(\ker T)$.

**Exercise 7.2.15** Define $T : \mathbf{P}_n \to \mathbb{R}$ by $T[p(x)] =$ the sum of all the coefficients of $p(x)$.

a. Use the dimension theorem to show that $\dim(\ker T) = n$.

b. Conclude that $\{x-1, x^2-1, \dots, x^n-1\}$ is a basis of $\ker T$.

**Exercise 7.2.16** Use the dimension theorem to prove Theorem 1.3.1: If $A$ is an $m \times n$ matrix with $m < n$, the system $A\mathbf{x} = \mathbf{0}$ of $m$ homogeneous equations in $n$ variables always has a nontrivial solution.

**Exercise 7.2.17** Let $B$ be an $n \times n$ matrix, and consider the subspaces $U = \{A \mid A \text{ in } \mathbf{M}_{mn},\ AB = 0\}$ and $V = \{AB \mid A \text{ in } \mathbf{M}_{mn}\}$. Show that $\dim U + \dim V = mn$.

**Exercise 7.2.18** Let $U$ and $V$ denote, respectively, the spaces of even and odd polynomials in $\mathbf{P}_n$. Show that $\dim U + \dim V = n+1$. [*Hint*: Consider $T : \mathbf{P}_n \to \mathbf{P}_n$ where $T[p(x)] = p(x) - p(-x)$.]

**Exercise 7.2.19** Show that every polynomial $f(x)$ in $\mathbf{P}_{n-1}$ can be written as $f(x) = p(x+1) - p(x)$ for some polynomial $p(x)$ in $\mathbf{P}_n$. [*Hint*: Define $T : \mathbf{P}_n \to \mathbf{P}_{n-1}$ by $T[p(x)] = p(x+1) - p(x)$.]

**Exercise 7.2.20** Let $U$ and $V$ denote the spaces of symmetric and skew-symmetric $n \times n$ matrices. Show that $\dim U + \dim V = n^2$.

**Exercise 7.2.21** Assume that $B$ in $\mathbf{M}_{nn}$ satisfies $B^k = 0$ for some $k \geq 1$. Show that every matrix in $\mathbf{M}_{nn}$ has the form $BA - A$ for some $A$ in $\mathbf{M}_{nn}$. [*Hint*: Show that $T : \mathbf{M}_{nn} \to \mathbf{M}_{nn}$ is linear and one-to-one where $T(A) = BA - A$ for each $A$.]

**Exercise 7.2.22** Fix a column $\mathbf{y} \neq \mathbf{0}$ in $\mathbb{R}^n$ and let $U = \{A \text{ in } \mathbf{M}_{nn} \mid A\mathbf{y} = \mathbf{0}\}$. Show that $\dim U = n(n-1)$.

**Exercise 7.2.23** If $B$ in $\mathbf{M}_{mn}$ has rank $r$, let $U = \{A \text{ in } \mathbf{M}_{nn} \mid BA = 0\}$ and $W = \{BA \mid A \text{ in } \mathbf{M}_{nn}\}$. Show that $\dim U = n(n-r)$ and $\dim W = nr$. [*Hint*: Show that $U$ consists of all matrices $A$ whose columns are in the null space of $B$. Use Example 7.2.7.]

**Exercise 7.2.24** Let $T : V \to V$ be a linear transformation where $\dim V = n$. If $\ker T \cap \operatorname{im} T = \{\mathbf{0}\}$, show that every vector $\mathbf{v}$ in $V$ can be written $\mathbf{v} = \mathbf{u} + \mathbf{w}$ for some $\mathbf{u}$ in $\ker T$ and $\mathbf{w}$ in $\operatorname{im} T$. [*Hint*: Choose bases $B \subseteq \ker T$ and $D \subseteq \operatorname{im} T$, and use Exercise 6.3.33.]

**Exercise 7.2.25** Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear operator of rank 1, where $\mathbb{R}^n$ is written as rows. Show that there exist numbers $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ such that $T(X) = XA$ for all rows $X$ in $\mathbb{R}^n$, where

$$A = \left[\begin{array}{cccc} a_1b_1 & a_1b_2 & \cdots & a_1b_n \\ a_2b_1 & a_2b_2 & \cdots & a_2b_n \\ \vdots & \vdots & & \vdots \\ a_nb_1 & a_nb_2 & \cdots & a_nb_n \end{array}\right]$$

[*Hint*: $\operatorname{im} T = \mathbb{R}\mathbf{w}$ for $\mathbf{w} = (b_1, \dots, b_n)$ in $\mathbb{R}^n$.]

**Exercise 7.2.26** Prove Theorem 7.2.5.

**Exercise 7.2.27** Let $T : V \to \mathbb{R}$ be a nonzero linear transformation, where $\dim V = n$. Show that there is a basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $V$ so that $T(r_1\mathbf{e}_1 + r_2\mathbf{e}_2 + \cdots + r_n\mathbf{e}_n) = r_1$.

**Exercise 7.2.28** Let $f \neq 0$ be a fixed polynomial of degree $m \geq 1$. If $p$ is any polynomial, recall that $(p \circ f)(x) = p[f(x)]$. Define $T_f : P_n \to P_{n+m}$ by $T_f(p) = p \circ f$.

a. Show that $T_f$ is linear.

b. Show that $T_f$ is one-to-one.

**Exercise 7.2.29** Let $U$ be a subspace of a finite dimensional vector space $V$.

a. Show that $U = \ker T$ for some linear operator $T : V \to V$.

b. Show that $U = \operatorname{im} S$ for some linear operator $S : V \to V$. [*Hint*: Theorem 6.4.1 and Theorem 7.1.3.]

**Exercise 7.2.30** Let $V$ and $W$ be finite dimensional vector spaces.

a. Show that $\dim W \leq \dim V$ if and only if there exists an onto linear transformation $T : V \to W$. [*Hint*: Theorem 6.4.1 and Theorem 7.1.3.]

b. Show that $\dim W \geq \dim V$ if and only if there exists a one-to-one linear transformation $T : V \to W$. [*Hint*: Theorem 6.4.1 and Theorem 7.1.3.]

**Exercise 7.2.31** Let $A$ and $B$ be $n \times n$ matrices, and assume that $AXB = 0$, $X \in \mathbf{M}_{nn}$, implies $X = 0$. Show that $A$ and $B$ are both invertible. [Hint: Dimension Theorem.]

**Example 7.3.8**

Define $T : \mathbb{R}^3 \to \mathbb{R}^3$ by $T(x, y, z) = (z, x, y)$. Show that $T^3 = 1_{\mathbb{R}^3}$, and hence find $T^{-1}$.

Solution. $T^2(x, y, z) = T[T(x, y, z)] = T(z, x, y) = (y, z, x)$. Hence

$$T^3(x, y, z) = T\left[T^2(x, y, z)\right] = T(y, z, x) = (x, y, z)$$

Since this holds for all $(x, y, z)$, it shows that $T^3 = 1_{\mathbb{R}^3}$, so $T(T^2) = 1_{\mathbb{R}^3} = (T^2)T$. Thus $T^{-1} = T^2$ by (2) of Theorem 7.3.5.

**Example 7.3.9**

Define $T : \mathbf{P}_n \to \mathbb{R}^{n+1}$ by $T(p) = (p(0), p(1), \dots, p(n))$ for all $p$ in $\mathbf{P}_n$. Show that $T^{-1}$ exists.

Solution. The verification that $T$ is linear is left to the reader. If $T(p) = 0$, then $p(k) = 0$ for $k = 0, 1, \dots, n$, so $p$ has $n+1$ distinct roots. Because $p$ has degree at most $n$, this implies that $p = 0$ is the zero polynomial (Theorem 6.5.4) and hence that $T$ is one-to-one. But $\dim \mathbf{P}_n = n+1 = \dim \mathbb{R}^{n+1}$, so this means that $T$ is also onto and hence is an isomorphism. Thus $T^{-1}$ exists by Theorem 7.3.5. Note that we have not given a description of the action of $T^{-1}$, we have merely shown that such a description exists. To give it explicitly requires some ingenuity; one method involves the Lagrange interpolation expansion (Theorem 6.5.3).

# Exercises for 7.3

**Exercise 7.3.1** Verify that each of the following is an isomorphism (Theorem 7.3.3 is useful).

a. $T : \mathbb{R}^3 \to \mathbb{R}^3$; $T(x, y, z) = (x+y, y+z, z+x)$

b. $T : \mathbb{R}^3 \to \mathbb{R}^3$; $T(x, y, z) = (x, x+y, x+y+z)$

c. $T : \mathbb{C} \to \mathbb{C}$; $T(z) = \overline{z}$

d. $T : \mathbf{M}_{mn} \to \mathbf{M}_{mn}$; $T(X) = UXV$, $U$ and $V$ invertible

e. $T : \mathbf{P}_1 \to \mathbb{R}^2$; $T[p(x)] = [p(0), p(1)]$

f. $T : V \to V$; $T(\mathbf{v}) = k\mathbf{v}$, $k \neq 0$ a fixed number, $V$ any vector space

g. $T : \mathbf{M}_{22} \to \mathbb{R}^4$; $T\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] = (a+b, d, c, a-b)$

h. $T : \mathbf{M}_{mn} \to \mathbf{M}_{nm}$; $T(A) = A^T$

**Exercise 7.3.2** Show that

$$\{a+bx+cx^2, a_1+b_1x+c_1x^2, a_2+b_2x+c_2x^2\}$$

is a basis of $\mathbf{P}_2$ if and only if $\{(a, b, c), (a_1, b_1, c_1), (a_2, b_2, c_2)\}$ is a basis of $\mathbb{R}^3$.

**Exercise 7.3.3** If $V$ is any vector space, let $V^n$ denote the space of all $n$-tuples $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$, where each $\mathbf{v}_i$ lies in $V$. (This is a vector space with component-wise operations; see Exercise 6.1.17.) If $C_j(A)$ denotes the $j$th column of the $m \times n$ matrix $A$, show that $T : \mathbf{M}_{mn} \to (\mathbb{R}^m)^n$ is an isomorphism if $T(A) = \left[\begin{array}{cccc} C_1(A) & C_2(A) & \cdots & C_n(A) \end{array}\right]$. (Here $\mathbb{R}^m$ consists of columns.)

**Exercise 7.3.4** In each case, compute the action of $ST$ and $TS$, and show that $ST \neq TS$.

a. $S : \mathbb{R}^2 \to \mathbb{R}^2$ with $S(x, y) = (y, x)$; $T : \mathbb{R}^2 \to \mathbb{R}^2$ with $T(x, y) = (x, 0)$

b. $S : \mathbb{R}^3 \to \mathbb{R}^3$ with $S(x, y, z) = (x, 0, z)$;
$T : \mathbb{R}^3 \to \mathbb{R}^3$ with $T(x, y, z) = (x+y, 0, y+z)$

c. $S : \mathbf{P}_2 \to \mathbf{P}_2$ with $S(p) = p(0) + p(1)x + p(2)x^2$;
$T : \mathbf{P}_2 \to \mathbf{P}_2$ with $T(a+bx+cx^2) = b+cx+ax^2$

d. $S : \mathbf{M}_{22} \to \mathbf{M}_{22}$ with $S\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$;
$T : \mathbf{M}_{22} \to \mathbf{M}_{22}$ with $T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & a \\ d & b \end{bmatrix}$

**Exercise 7.3.5** In each case, show that the linear transformation $T$ satisfies $T^2 = T$.

a. $T : \mathbb{R}^4 \to \mathbb{R}^4$; $T(x, y, z, w) = (x, 0, z, 0)$

b. $T : \mathbb{R}^2 \to \mathbb{R}^2$; $T(x, y) = (x+y, 0)$

c. $T : \mathbf{P}_2 \to \mathbf{P}_2$;
$T(a+bx+cx^2) = (a+b-c)+cx+cx^2$

d. $T : \mathbf{M}_{22} \to \mathbf{M}_{22}$;
$T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{2}\begin{bmatrix} a+c & b+d \\ a+c & b+d \end{bmatrix}$

**Exercise 7.3.6** Determine whether each of the following transformations $T$ has an inverse and, if so, determine the action of $T^{-1}$.

a. $T : \mathbb{R}^3 \to \mathbb{R}^3$;
$T(x, y, z) = (x+y, y+z, z+x)$

b. $T : \mathbb{R}^4 \to \mathbb{R}^4$;
$T(x, y, z, t) = (x+y, y+z, z+t, t+x)$

c. $T : \mathbf{M}_{22} \to \mathbf{M}_{22}$;
$T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a-c & b-d \\ 2a-c & 2b-d \end{bmatrix}$

d. $T : \mathbf{M}_{22} \to \mathbf{M}_{22}$;
$T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+2c & b+2d \\ 3c-a & 3d-b \end{bmatrix}$

e. $T : \mathbf{P}_2 \to \mathbb{R}^3$; $T(a+bx+cx^2) = (a-c, 2b, a+c)$

f. $T : \mathbf{P}_2 \to \mathbb{R}^3$; $T(p) = [p(0), p(1), p(-1)]$

**Exercise 7.3.7** In each case, show that $T$ is self-inverse, that is: $T^{-1} = T$.

a. $T : \mathbb{R}^4 \to \mathbb{R}^4$; $T(x, y, z, w) = (x, -y, -z, w)$

b. $T : \mathbb{R}^2 \to \mathbb{R}^2$; $T(x, y) = (ky-x, y)$, $k$ any fixed number

c. $T : \mathbf{P}_n \to \mathbf{P}_n$; $T(p(x)) = p(3-x)$

d. $T : \mathbf{M}_{22} \to \mathbf{M}_{22}$; $T(X) = AX$ where $A = \frac{1}{4}\begin{bmatrix} 5 & -3 \\ 3 & -5 \end{bmatrix}$

**Exercise 7.3.8** In each case, show that $T^6 = 1_{R^4}$ and so determine $T^{-1}$.

a. $T : \mathbb{R}^4 \to \mathbb{R}^4$; $T(x, y, z, w) = (-x, z, w, y)$

b. $T : \mathbb{R}^4 \to \mathbb{R}^4$; $T(x, y, z, w) = (-y, x-y, z, -w)$

**Exercise 7.3.9** In each case, show that $T$ is an isomorphism by defining $T^{-1}$ explicitly.

a. $T : \mathbf{P}_n \to \mathbf{P}_n$ is given by $T[p(x)] = p(x+1)$.

b. $T : \mathbf{M}_{nn} \to \mathbf{M}_{nn}$ is given by $T(A) = UA$ where $U$ is invertible in $\mathbf{M}_{nn}$.

**Exercise 7.3.10** Given linear transformations $V \xrightarrow{T} W \xrightarrow{S} U$:

a. If $S$ and $T$ are both one-to-one, show that $ST$ is one-to-one.

b. If $S$ and $T$ are both onto, show that $ST$ is onto.

**Exercise 7.3.11** Let $T : V \to W$ be a linear transformation.

a. If $T$ is one-to-one and $TR = TR_1$ for transformations $R$ and $R_1 : U \to V$, show that $R = R_1$.

b. If $T$ is onto and $ST = S_1T$ for transformations $S$ and $S_1 : W \to U$, show that $S = S_1$.

**Exercise 7.3.12** Consider the linear transformations $V \xrightarrow{T} W \xrightarrow{R} U$.

a. Show that $\ker T \subseteq \ker RT$.

b. Show that $\operatorname{im} RT \subseteq \operatorname{im} R$.

**Exercise 7.3.13** Let $V \xrightarrow{T} U \xrightarrow{S} W$ be linear transformations.

a. If $ST$ is one-to-one, show that $T$ is one-to-one and that $\dim V \leq \dim U$.

b. If $ST$ is onto, show that $S$ is onto and that $\dim W \leq \dim U$.

**Exercise 7.3.14** Let $T : V \to V$ be a linear transformation. Show that $T^2 = 1_V$ if and only if $T$ is invertible and $T = T^{-1}$.

**Exercise 7.3.15** Let $N$ be a nilpotent $n \times n$ matrix (that is, $N^k = 0$ for some $k$). Show that $T : \mathbf{M}_{nm} \to \mathbf{M}_{nm}$ is an isomorphism if $T(X) = X - NX$. [*Hint*: If $X$ is in $\ker T$, show that $X = NX = N^2X = \cdots$. Then use Theorem 7.3.3.]

**Exercise 7.3.16** Let $T : V \to W$ be a linear transformation, and let $\{\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{e}_{r+1}, \ldots, \mathbf{e}_n\}$ be any basis of $V$ such that $\{\mathbf{e}_{r+1}, \ldots, \mathbf{e}_n\}$ is a basis of $\ker T$. Show that $\operatorname{im} T \cong \operatorname{span}\{\mathbf{e}_1, \ldots, \mathbf{e}_r\}$. [*Hint*: See Theorem 7.2.5.]

**Exercise 7.3.17** Is every isomorphism $T : \mathbf{M}_{22} \to \mathbf{M}_{22}$ given by an invertible matrix $U$ such that $T(X) = UX$ for all $X$ in $\mathbf{M}_{22}$? Prove your answer.

**Exercise 7.3.18** Let $\mathbf{D}_n$ denote the space of all functions $f$ from $\{1, 2, \ldots, n\}$ to $\mathbb{R}$ (see Exercise 6.3.35). If $T : \mathbf{D}_n \to \mathbb{R}^n$ is defined by

$$T(f) = (f(1), f(2), \ldots, f(n)),$$

show that $T$ is an isomorphism.

**Exercise 7.3.19**

a. Let $V$ be the vector space of Exercise 6.1.3. Find an isomorphism $T : V \to \mathbb{R}^1$.

b. Let $V$ be the vector space of Exercise 6.1.4. Find an isomorphism $T : V \to \mathbb{R}^2$.

**Exercise 7.3.20** Let $V \xrightarrow{T} W \xrightarrow{S} V$ be linear transformations such that $ST = 1_V$. If $\dim V = \dim W = n$, show that $S = T^{-1}$ and $T = S^{-1}$. [*Hint*: Exercise 7.3.13 and Theorem 7.3.3, Theorem 7.3.4, and Theorem 7.3.5.]

**Exercise 7.3.21** Let $V \xrightarrow{T} W \xrightarrow{S} V$ be functions such that $TS = 1_W$ and $ST = 1_V$. If $T$ is linear, show that $S$ is also linear.

**Exercise 7.3.22** Let $A$ and $B$ be matrices of size $p \times m$ and $n \times q$. Assume that $mn = pq$. Define $R : \mathbf{M}_{mn} \to \mathbf{M}_{pq}$ by $R(X) = AXB$.

a. Show that $\mathbf{M}_{mn} \cong \mathbf{M}_{pq}$ by comparing dimensions.

b. Show that $R$ is a linear transformation.

c. Show that if $R$ is an isomorphism, then $m = p$ and $n = q$. [*Hint*: Show that $T : \mathbf{M}_{mn} \to \mathbf{M}_{pn}$ given by $T(X) = AX$ and $S : \mathbf{M}_{mn} \to \mathbf{M}_{mq}$ given by $S(X) = XB$ are both one-to-one, and use the dimension theorem.]

**Exercise 7.3.23** Let $T : V \to V$ be a linear transformation such that $T^2 = 0$ is the zero transformation.

a. If $V \neq \{\mathbf{0}\}$, show that $T$ cannot be invertible.

b. If $R : V \to V$ is defined by $R(\mathbf{v}) = \mathbf{v} + T(\mathbf{v})$ for all $\mathbf{v}$ in $V$, show that $R$ is linear and invertible.

**Exercise 7.3.24** Let $V$ consist of all sequences $[x_0, x_1, x_2, \ldots)$ of numbers, and define vector operations

$$[x_o, x_1, \ldots) + [y_0, y_1, \ldots) = [x_0 + y_0, x_1 + y_1, \ldots)$$
$$r[x_0, x_1, \ldots) = [rx_0, rx_1, \ldots)$$

a. Show that $V$ is a vector space of infinite dimension.

b. Define $T : V \to V$ and $S : V \to V$ by $T[x_0, x_1, \ldots) = [x_1, x_2, \ldots)$ and $S[x_0, x_1, \ldots) = [0, x_0, x_1, \ldots)$. Show that $TS = 1_V$, so $TS$ is one-to-one and onto, but that $T$ is not one-to-one and $S$ is not onto.

**Exercise 7.3.25** Prove (1) and (2) of Theorem 7.3.4.

**Exercise 7.3.26** Define $T : \mathbf{P}_n \to \mathbf{P}_n$ by $T(p) = p(x) + xp'(x)$ for all $p$ in $\mathbf{P}_n$.

a. Show that $T$ is linear.

b. Show that $\ker T = \{\mathbf{0}\}$ and conclude that $T$ is an isomorphism. [*Hint*: Write $p(x) = a_0 + a_1x + \cdots + a_nx^n$ and compare coefficients if $p(x) = -xp'(x)$.]

c. Conclude that each $q(x)$ in $\mathbf{P}_n$ has the form $q(x) = p(x) + xp'(x)$ for some unique polynomial $p(x)$.

d. Does this remain valid if $T$ is defined by $T[p(x)] = p(x) - xp'(x)$? Explain.

**Exercise 7.3.27** Let $T : V \to W$ be a linear transformation, where $V$ and $W$ are finite dimensional.

a. Show that $T$ is one-to-one if and only if there exists a linear transformation $S : W \to V$ with $ST = 1_V$. [*Hint*: If $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis of $V$ and $T$ is one-to-one, show that $W$ has a basis $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_n), \mathbf{f}_{n+1}, \dots, \mathbf{f}_{n+k}\}$ and use Theorem 7.1.2 and Theorem 7.1.3.]

b. Show that $T$ is onto if and only if there exists a linear transformation $S : W \to V$ with $TS = 1_W$. [*Hint*: Let $\{\mathbf{e}_1, \dots, \mathbf{e}_r, \dots, \mathbf{e}_n\}$ be a basis of $V$ such that $\{\mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ is a basis of $\ker T$. Use Theorem 7.2.5, Theorem 7.1.2 and Theorem 7.1.3.]

**Exercise 7.3.28** Let $S$ and $T$ be linear transformations $V \to W$, where $\dim V = n$ and $\dim W = m$.

a. Show that $\ker S = \ker T$ if and only if $T = RS$ for some isomorphism $R : W \to W$. [*Hint*: Let $\{\mathbf{e}_1, \dots, \mathbf{e}_r, \dots, \mathbf{e}_n\}$ be a basis of $V$ such that $\{\mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ is a basis of $\ker S = \ker T$. Use Theorem 7.2.5 to extend $\{S(\mathbf{e}_1), \dots, S(\mathbf{e}_r)\}$ and $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_r)\}$ to bases of $W$.]

b. Show that $\operatorname{im} S = \operatorname{im} T$ if and only if $T = SR$ for some isomorphism $R : V \to V$. [*Hint*: Show that $\dim(\ker S) = \dim(\ker T)$ and choose bases $\{\mathbf{e}_1, \dots, \mathbf{e}_r, \dots, \mathbf{e}_n\}$ and $\{\mathbf{f}_1, \dots, \mathbf{f}_r, \dots, \mathbf{f}_n\}$ of $V$ where $\{\mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ and $\{\mathbf{f}_{r+1}, \dots, \mathbf{f}_n\}$ are bases of $\ker S$ and $\ker T$, respectively. If $1 \le i \le r$, show that $S(\mathbf{e}_i) = T(\mathbf{g}_i)$ for some $\mathbf{g}_i$ in $V$, and prove that $\{\mathbf{g}_1, \dots, \mathbf{g}_r, \mathbf{f}_{r+1}, \dots, \mathbf{f}_n\}$ is a basis of $V$.]

**Exercise 7.3.29** If $T : V \to V$ is a linear transformation where $\dim V = n$, show that $TST = T$ for some isomorphism $S : V \to V$. [*Hint*: Let $\{\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{e}_{r+1}, \dots, \mathbf{e}_n\}$ be as in Theorem 7.2.5. Extend $\{T(\mathbf{e}_1), \dots, T(\mathbf{e}_r)\}$ to a basis of $V$, and use Theorem 7.3.1, Theorem 7.1.2 and Theorem 7.1.3.]

**Exercise 7.3.30** Let $A$ and $B$ denote $m \times n$ matrices. In each case show that (1) and (2) are equivalent.

a. (1) $A$ and $B$ have the same null space. (2) $B = PA$ for some invertible $m \times m$ matrix $P$.

b. (1) $A$ and $B$ have the same range. (2) $B = AQ$ for some invertible $n \times n$ matrix $Q$.

[*Hint*: Use Exercise 7.3.28.]

# 7.4 A Theorem about Differential Equations

Differential equations are instrumental in solving a variety of problems throughout science, social science, and engineering. In this brief section, we will see that the set of solutions of a linear differential equation (with constant coefficients) is a vector space and we will calculate its dimension. The proof is pure linear algebra, although the applications are primarily in analysis. However, a key result (Lemma 7.4.3 below) can be applied much more widely.

We denote the derivative of a function $f : \mathbb{R} \to \mathbb{R}$ by $f'$, and $f$ will be called **differentiable** if it can be differentiated any number of times. If $f$ is a differentiable function, the $n$th derivative $f^{(n)}$ of $f$ is the result of differentiating $n$ times. Thus $f^{(0)} = f$, $f^{(1)} = f'$, $f^{(2)} = f^{(1)\prime}$, $\dots$, and in general $f^{(n+1)} = f^{(n)\prime}$ for each $n \ge 0$. For small values of $n$ these are often written as $f, f', f'', f''', \dots$.

If $a$, $b$, and $c$ are numbers, the differential equations

$$f'' - af' - bf = 0 \quad \text{or} \quad f''' - af'' - bf' - cf = 0$$

are said to be of **second order** and **third-order**, respectively. In general, an equation

$$f^{(n)} - a_{n-1}f^{(n-1)} - a_{n-2}f^{(n-2)} - \cdots - a_2 f^{(2)} - a_1 f^{(1)} - a_0 f^{(0)} = 0, \; a_i \text{ in } \mathbb{R} \tag{7.3}$$